








Modelling Financial Markets

Chapter 5: Models of Conditional Volatility



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Can We Forecast Tomorrow's Risk?



Robert F. Engle. **Photo: Jon Demske (2022); CC BY-SA 4.0; Wikimedia Commons**

"These are mean zero, serially uncorrelated processes with nonconstant variances conditional on the past, but constant unconditional variances. For such processes, the recent past gives information about the one-period forecast variance."

— *Engle (1982)*

- S&P 500: the lag-1 autocorrelation is -0.099 for daily returns and 0.313 for squared returns
  - the sign of a return is nearly unpredictable, its size is not
- ARCH tests (autoregressive conditional heteroskedasticity) reject a constant variance in all six markets of the chapter
  - S&P 500: $Q(10)$ on r_t^2 is 6 135, against a 5% critical value of 18.3
- GARCH(1,1), the generalised ARCH model, with Student-t innovations: persistence $\hat{\alpha} + \hat{\beta} = 0.9948$ and a half-life of 132 days for the S&P 500
  - for Bitcoin and EUR/RON the estimate hits the bound $\alpha + \beta = 1$: in the model, shocks never fade
- Out of sample, 2016–2026 (2 693 one-day S&P 500 forecasts): GARCH(1,1)-t has a lower QLIKE (quasi-likelihood) loss than EWMA (exponentially weighted moving average) and than the mean of r^2 over the last 20 days
  - Diebold–Mariano $t = 3.47$ and 4.10

▸ Central chart: Out-of-Sample Design and Forecast Paths, 2016–2026

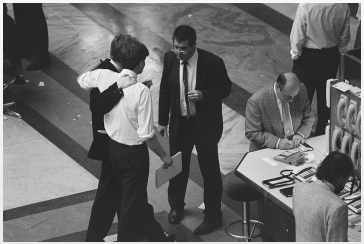
The Idea of the Chapter

- ARCH means AutoRegressive Conditional Heteroskedasticity: variance depends on past shocks
- Core topics
  - Clustering and the ARCH tests; the GARCH recursion, persistence and half-life; estimation with Student-t innovations; forecasts; asymmetry (GJR, EGARCH); residual diagnostics; forecast evaluation with QLIKE, Mincer–Zarnowitz and Diebold–Mariano
- Inference beyond the basic model
  - QMLE asymptotics, the delta method near $\alpha + \beta = 1$, tests on the boundary, corrected residual tests, Giacomini–White and the Model Confidence Set
- Two research sections: a quasi score-driven volatility model and an AI-assisted discovery loop
- Detailed examples and advanced topics are in Further Topics, after the Conclusions
  - the footer button leads there; the Back button returns to the starting slide

What you should be able to do

- derive the long-run variance, the kurtosis and the half-life of GARCH(1,1)
- compute multi-day variance forecasts and LR tests on the boundary
- interpret GARCH, GJR and EGARCH estimates in six markets
- interpret out-of-sample comparisons with QLIKE, DM and the Model Confidence Set
- Further Topics and Appendix: reading for familiarity

Volatility Comes in Bursts

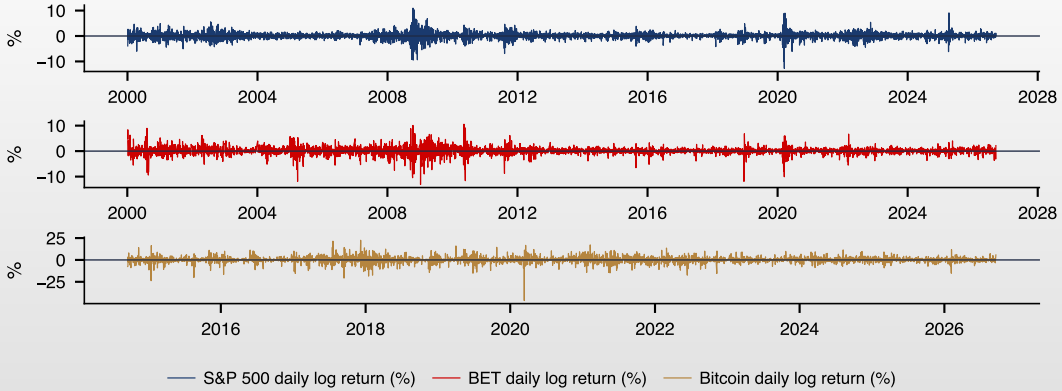


Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- Crashes are not confined to isolated days: large moves are followed by large moves
  - "Large changes tend to be followed by large changes, of either sign" Mandelbrot (1963)
- This **volatility clustering** is stylised fact 6 of Chapter 1
- Volatility enters every risk number
  - Value at Risk (Chapter 7)
  - Portfolio weights (Chapter 4)
  - Option prices
- Beyond the forecast: can we conduct valid inference on it?

Volatility Clusters in Every Market



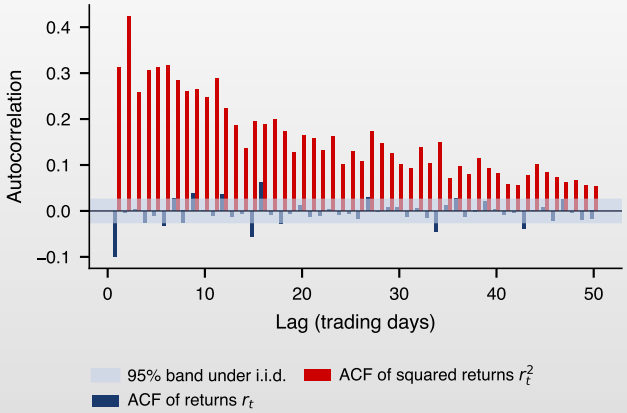
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Calendar and Volatility Episodes

- Daily log returns $r_t = 100\,(\ln P_t - \ln P_{t-1})$ in %, closes up to 18 September 2026
  - P_t: closing price (index level) on day t
- Six series, chosen for different market regimes
  - The S&P 500 (the global benchmark) and the BET (the local market), from 2000
  - BET-TR (the BET with reinvested dividends), from 2014
  - Bitcoin (BTC), from 17 September 2014, 7 days a week
  - EUR/RON (the BNR reference rate, a managed float), from 2005
  - Gold (XAU/USD), a safe-haven asset, from 2000
- Equity indices and gold: weekdays only
  - A heuristic rule: a row that repeats the previous close and shows no trading (reported zero volume, open = high = low = close, or a copy of the previous row) is treated as a provider fill
  - Such rows are dropped, with the reason kept, and checked against the exchange calendar; a genuine unchanged close (a zero return) is kept
  - A missing volume is not treated as zero volume
  - Days dropped: 0 for the S&P 500, the BET and BET-TR, 7 for gold
- In the chart (S&P 500, BET, Bitcoin), quiet years alternate with turbulent months
  - 2008, March 2020, April 2025

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The Sign Is Nearly Unpredictable, the Size Is Not



- Sample autocorrelation at lag k: $\hat{\rho}_k = \widehat{\mathrm{Corr}}(x_t, x_{t-k})$, computed for $x_t = r_t$ and $x_t = r_t^2$
  - Band $\pm 1.96/\sqrt{n}$ (n: number of returns): 95% range of $\hat{\rho}_k$ if the series were i.i.d.
- S&P 500, lag 1: -0.099 for r_t, 0.313 for r_t^2
- For r_t, 15 of 50 autocorrelations lie outside ± 0.024, but all are small
  - For r_t^2, all 50 do
- Heavy tails too: excess kurtosis (kurtosis minus 3) 10.7 (S&P 500), 11.1 (BET), 12.0 (Bitcoin)
  - The kurtosis requires a finite fourth moment, which Chapter 1 questions
- Object to model: $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$, with $\mathcal{F}_{t-1}$ the information at the close of day $t-1$

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Testing for ARCH Effects (1/2): the Two Tests

- H_0: no clustering, $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$ constant
  - Both tests look for autocorrelation in squared returns
- **McLeod–Li** McLeod & Li (1983): the Ljung–Box statistic applied to r_t^2
  - $Q(m) = n(n+2)\sum_{k=1}^{m} \dfrac{\hat{\rho}_k^2}{n-k}$
  - $\hat{\rho}_k$: autocorrelation of r_t^2 at lag k; n: sample size; $m = 10$ lags (two trading weeks)
  - Measures the joint size of the first m autocorrelations; χ^2_{10} under H_0; reject if $Q(10) > 18.3$ (5%)
- **Engle: Lagrange Multiplier (LM) test** Engle (1982)
  - Regress $\hat{\varepsilon}_t^2$ on a constant and $\hat{\varepsilon}_{t-1}^2, \ldots, \hat{\varepsilon}_{t-5}^2$ (5 lags: one trading week)
  - $\hat{\varepsilon}_t = r_t - \bar{r}$: demeaned return, $\bar{r}$: sample mean
  - $LM = nR^2$, with R^2 the coefficient of determination of this regression; H_0: all five slopes are 0
  - χ^2_5 under H_0; reject if $LM > 11.1$ (5%); a large LM means past squared shocks predict today's

Testing for ARCH Effects (2/2): Results

- Every statistic is far above its critical value (all $p < 0.001$)
  - Clustering in equities, Bitcoin, EUR/RON and gold
  - The χ^2 p-values require at least a finite fourth moment of r_t, which the Hill estimates of Chapter 1 put in doubt
- The weakest effect, by $\hat{\rho}_1(r^2)$, is for gold (0.110) and Bitcoin (0.137)
- Q and LM grow with n, so they are not comparable across series of different lengths
  - One-sided score tests: Appendix
- What the result supports: description

Series	$Q(10)$ on r^2	LM(5)	$\hat{\rho}_1(r^2)$
S&P 500	6 135	1 685	0.313
BET	2 440	982	0.330
BET-TR	714	371	0.171
Bitcoin	213	115	0.137
EUR/RON	2 420	871	0.342
Gold	745	331	0.110

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The ARCH Model of Engle (1982)

- Setting: $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, innovations z_t i.i.d. with mean 0 and variance 1
  - μ: mean return; ε_t: shock (unexpected part of the return); σ_t: conditional volatility
  - Conditional variance $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$; unconditional variance $\sigma^2 = \mathbb{E}\sigma_t^2$
- **ARCH**(q): today's variance is a constant plus a weighted sum of the last q squared shocks Engle (1982)
  - $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2$
  - q: number of lags; ω: variance floor; α_i: weight of the shock i days ago
  - $\omega > 0$, $\alpha_i \geq 0$: the variance is always positive
- A large shock yesterday (ε_{t-1}^2 large) raises today's variance: clustering by construction
- ARCH(1): unconditional variance $\sigma^2 = \omega/(1-\alpha_1)$, finite if $\alpha_1 < 1$
- Engle received the 2003 Nobel Prize for ARCH (history: Appendix)

From ARCH to GARCH (Bollerslev, 1986)

- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
  - The variance also depends on its own past
- Problem with ARCH: clustering in real data lasts for months, so q must be large (many parameters)
- **GARCH**(1,1) Bollerslev (1986): $\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2$
  - $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$; α: reaction to news; β: memory of past variance
- Substituting backwards: $\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j\geq 1} \beta^{j-1} \varepsilon_{t-j}^2$
  - Valid for $\beta < 1$; j: age of the shock in days, weight $\alpha\beta^{j-1}$
  - An ARCH(∞) with geometrically declining weights, using only three parameters
- GARCH(p,q) adds more lags (p lags of σ^2, q lags of ε^2)
  - In practice GARCH(1,1) is the workhorse Hansen & Lunde (2005)

Model Labels and Innovations

- The model name describes the variance dynamics
  - The suffix describes the law of the standardised innovation
- GARCH-N: GARCH with innovations from the Normal distribution
- GARCH-t: GARCH with Student-t innovations, which allow heavier tails
- GJR-t (or GJR-GARCH-t)
  - Asymmetric Glosten–Jagannathan–Runkle variance (Asymmetry section) with Student-t innovations

GARCH(1,1) as an ARMA Model for Squared Shocks

- Define $u_t = \varepsilon_t^2 - \sigma_t^2$, the surprise in the squared shock ($\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] = 0$)
- Then $\varepsilon_t^2 = \omega + (\alpha + \beta)\,\varepsilon_{t-1}^2 + u_t - \beta\, u_{t-1}$
  - Obtained by adding u_t to both sides of the GARCH recursion and writing $\sigma_{t-1}^2 = \varepsilon_{t-1}^2 - u_{t-1}$
  - An AutoRegressive Moving Average model, ARMA(1,1), in ε_t^2 with autoregressive coefficient $\alpha + \beta$ and moving-average coefficient $-\beta$
- Consequence: the autocorrelations of ε_t^2 decay geometrically, $\rho_k = \rho_1(\alpha + \beta)^{k-1}$
  - $\rho_k = \mathrm{Corr}(\varepsilon_t^2, \varepsilon_{t-k}^2)$: autocorrelation of squared shocks at lag k
  - $\rho_1 = \alpha(1 - \beta^2 - \alpha\beta)/(1 - \beta^2 - 2\alpha\beta)$ if $\mathbb{E}\varepsilon_t^4 < \infty$ (Seminar 5, A6)
  - S&P 500, GARCH-N by Gaussian QMLE ($\hat{\alpha} = 0.120$, $\hat{\beta} = 0.860$): $\rho_1 = 0.356$ and $\rho_{10} = 0.298$, against the sample values $\hat{\rho}_1 = 0.313$ and $\hat{\rho}_{10} = 0.248$
- $\alpha + \beta$ is therefore called the **persistence** of volatility

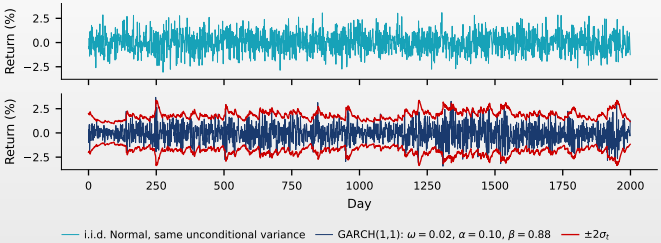
Stationarity and the Unconditional Variance

- **Covariance stationarity**: $\alpha + \beta < 1$; then
  - $\sigma^2 = \mathbb{E}[\sigma_t^2] = \dfrac{\omega}{1-\alpha-\beta}$
- Derivation: take expectations in the recursion, $\mathbb{E}\sigma_t^2 = \omega + (\alpha+\beta)\,\mathbb{E}\sigma_{t-1}^2$, and set $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2$
- **IGARCH** (integrated GARCH): $\alpha + \beta = 1$
  - The unconditional variance is infinite
  - Yet, if $\alpha > 0$ and z_t^2 is non-degenerate, the process is still **strictly** stationary: the condition $\mathbb{E}\ln(\alpha z_t^2 + \beta) < 0$ Nelson (1990) holds by Jensen, since $\mathbb{E}\ln(\alpha z_t^2 + \beta) < \ln(\alpha + \beta) = 0$
- Notation: σ^2 is the model's long-run variance (Seminar 5 writes $\bar{\sigma}^2$)
  - $\hat{\sigma}^2 = \hat{\omega}/(1-\hat{\alpha}-\hat{\beta})$: its estimate
  - s^2: the sample variance of r_t
- Variance targeting: fix $\omega = s^2(1-\alpha-\beta)$, so that the long-run variance equals s^2 and only α, β are estimated

Fourth Moment and Kurtosis

- With Normal z_t, the kurtosis of ε_t is $K = \dfrac{3\left[1-(\alpha+\beta)^2\right]}{1-(\alpha+\beta)^2-2\alpha^2}$
  - $K = \mathbb{E}\varepsilon_t^4/(\mathbb{E}\varepsilon_t^2)^2$: 3 for the Normal distribution, above 3 for heavy tails
  - Finite only if $(\alpha+\beta)^2 + 2\alpha^2 < 1$ (Seminar 5, A4)
  - ARCH(1) is the case $\beta = 0$: $K = 3(1-\alpha^2)/(1-3\alpha^2)$, finite only if $\alpha < 1/\sqrt{3}$: heavy tails even with Normal innovations
- S&P 500, GARCH-N, 2000–2026: $\hat{\alpha} = 0.120487$, $\hat{\beta} = 0.860062$
  - $(\hat{\alpha}+\hat{\beta})^2 + 2\hat{\alpha}^2 = 0.990509$: just below 1, so $K = 12.2$; sample kurtosis of the returns: 13.7 (excess kurtosis 10.7)
- Near the moment boundary, K is very sensitive to rounding the estimates (Appendix)
- Clustering alone generates much of the fat tails
  - Heavy-tailed innovations z_t may contribute further; the gap between 12.2 and 13.7 is not an identified decomposition

Simulation: GARCH Produces Clustering



- Top: i.i.d. Normal returns
- Bottom: GARCH(1,1) with $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$
  - The same unconditional variance (1), and the bands $\pm 2\sigma_t$
- Quiet stretches and bursts appear only in the GARCH path
- Normal innovations, yet fat-tailed returns
  - A mixture of Normal distributions with varying variances has heavy tails
  - Kurtosis on a long path: ▸ Appendix

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Persistence and Half-Life

- Multi-step forecast: $\mathbb{E}[\sigma^2_{t+h} \mid \mathcal{F}_t] = \sigma^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \sigma^2)$
  - h: forecast horizon in days; σ^2_{t+1}: one-day-ahead variance, known at t
  - The gap to the long-run variance shrinks by the factor $\alpha+\beta$ each day
- **Half-life**: number of days after which half of a variance shock has gone
  - $(\alpha+\beta)^h = 0.5 \;\Rightarrow\; h_{1/2} = \dfrac{\ln 0.5}{\ln(\alpha+\beta)}$
- Half-life: 13.5 days for $\alpha+\beta = 0.95$
  - 69 days for 0.99
  - 138 days for 0.995
- Small changes in $\alpha+\beta$ near 1 imply very different half-lives
  - Estimation uncertainty matters (next section; Seminar 5, A2 and B4)

A Toy Check: Multi-Day Risk After a Turbulent Day

- Toy GARCH(1,1), not fitted to any data: $\omega = 0.04$, $\alpha = 0.08$, $\beta = 0.90$
  - Persistence $p = \alpha + \beta = 0.98$; long-run variance $\sigma^2 = 0.04/0.02 = 2$; half-life $\ln 0.5 / \ln 0.98 \approx 34.3$ days
- A turbulent day: $\sigma^2_{t+1} = 4$, twice the long-run level
  - Day 2: $\mathbb{E}[\sigma^2_{t+2} \mid \mathcal{F}_t] = 2 + 0.98 \times (4 - 2) = 3.96$: the gap to σ^2 shrinks by 2% a day
- H-day variance: a geometric sum
  - Daily shocks are uncorrelated
  - So the variance of their sum is the sum of the daily variance forecasts
  - $V_H = \sum_{h=1}^{H} \mathbb{E}[\sigma^2_{t+h} \mid \mathcal{F}_t] = H\sigma^2 + (\sigma^2_{t+1} - \sigma^2)\,\dfrac{1 - p^H}{1 - p}$
  - $H = 5$: $V_5 = 10 + 2 \times (1 - 0.98^5)/0.02 = 10 + 2 \times 4.804 = 19.61$; the square-root-of-time rule, $V_H = H\,\sigma^2_{t+1}$ (H-day volatility is $\sqrt{H}$ times the one-day volatility), gives $5 \times 4 = 20$, too high
- Your turn: redo $H = 5$ after a calm day, $\sigma^2_{t+1} = 1$
  - Answer: $V_5 = 10 - 4.804 = 5.20 > 5$
  - So the rule is now too low, as in the S&P 500 case of the Forecasts section

EWMA and IGARCH: No Mean Reversion

- **Exponentially Weighted Moving Average (EWMA)** J.P. Morgan/Reuters (1996)
  - $\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,r_{t-1}^2$
  - $\lambda = 0.94$ for daily data and zero-mean returns
  - $\lambda \in (0,1)$: decay factor
  - Weights $(1-\lambda)\lambda^{j-1}$ on past squared returns: the most recent days count most
  - EWMA is an IGARCH(1,1) with $\omega = 0$, $\alpha = 1-\lambda$, $\beta = \lambda$: no parameter to estimate
- Multi-step forecasts at the close of day t differ
  - $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_t]$: expectation given the information at t
  - EWMA: flat, $\mathbb{E}_t\sigma_{t+h}^2 = \sigma_{t+1}^2$ for every h
  - IGARCH with $\omega > 0$: rising, $\mathbb{E}_t\sigma_{t+h}^2 = \sigma_{t+1}^2 + (h-1)\,\omega$
  - Stationary GARCH: pulled towards σ^2 at the rate $\alpha + \beta$
- EWMA serves as a benchmark in the forecast comparison

Maximum Likelihood

- Maximum Likelihood Estimation (MLE) of $\theta = (\mu, \omega, \alpha, \beta)$
  - Given θ, the recursion produces $\sigma_1^2(\theta), \ldots, \sigma_n^2(\theta)$
- Normal log-likelihood: $\ell(\theta) = -\frac{1}{2} \sum_{t=1}^{n} \left[\ln 2\pi + \ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2 \right]$
  - $\varepsilon_t = r_t - \mu$
  - Each day contributes the density of ε_t given the past: the likelihood factorises in time
- No closed form: numerical optimisation under $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta \leq 1$
  - The `arch` package imposes $\alpha + \beta \leq 1$, a closed set: an estimate can sit on the bound (Bitcoin and EUR/RON in the six-market table)
- Start-up: σ_1^2 is a backcast from the first squared residuals, whose effect fades at rate β
- Scale: returns in %, so that the variance is of order 1
  - EUR/RON has a daily standard deviation of about 0.28%, so it is fitted on $10\, r_t$; on unscaled returns the optimiser stops at a local optimum

Heavy-Tailed Innovations

- **Student-t** Bollerslev (1987): $z_t = \sqrt{(\nu-2)/\nu}\, t_\nu$, unit variance for $\nu > 2$
  - t_ν: a standard Student-t variable, variance $\nu/(\nu-2)$; the factor rescales it to variance 1
  - ν (degrees of freedom): the smaller ν, the heavier the tails; the fourth moment exists only for $\nu > 4$
- **Skewed-t** Hansen (1994): adds an asymmetry parameter $\lambda \in (-1, 1)$
  - $\lambda < 0$: the left tail is longer (large losses more likely than large gains)
- **Generalised Error Distribution (GED)** Nelson (1991): density $\propto \exp(-\frac{1}{2}|z/\lambda_\nu|^\nu)$
  - $\nu > 0$: shape parameter; λ_ν: scale constant
  - $\nu = 2$: the Normal distribution; $\nu = 1$: Laplace; $\nu < 2$: heavier tails (λ_ν gives unit variance)

The Student-t Likelihood

- Day t's contribution to the GARCH-t log-likelihood, the chapter's baseline model ($\varepsilon_t = r_t - \mu$)

$$\ell_t = \ln \Gamma\left(\tfrac{\nu+1}{2}\right) - \ln \Gamma\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2} \ln[\pi(\nu-2)] - \tfrac{1}{2} \ln \sigma_t^2 - \tfrac{\nu+1}{2} \ln\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right)$$

- Γ: the gamma function
  - $\nu > 2$: degrees of freedom of the unit-variance innovation
- As $\nu \to \infty$, ℓ_t becomes the Normal contribution of the Maximum Likelihood slide
- Why Student-t: a single extra parameter (ν) and symmetric heavy tails

Gaussian QMLE: Definition and Consistency

- **Gaussian QMLE** (Quasi-Maximum Likelihood Estimator) Bollerslev & Wooldridge (1992)
  - Maximise the Normal likelihood even if z_t is not Normal
  - The Normal $\ell(\theta)$ is used as a criterion, not as a claim about the law of z_t
  - θ_0: true parameter value; $\hat{\theta}$: the estimate, the maximiser of ℓ
- **Consistency** (GARCH(1,1)): $\hat{\theta} \to \theta_0$ as $n \to \infty$ Berkes, Horváth & Kokoszka (2003), Francq & Zakoïan (2004)
  - Strict stationarity: $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$
  - $\alpha_0 > 0$ (otherwise β is not identified) and $\beta_0 < 1$
  - θ_0 in a compact (closed and bounded) parameter set; z_t^2 non-degenerate (not a constant)
  - No moment condition on ε_t: consistency holds even for IGARCH

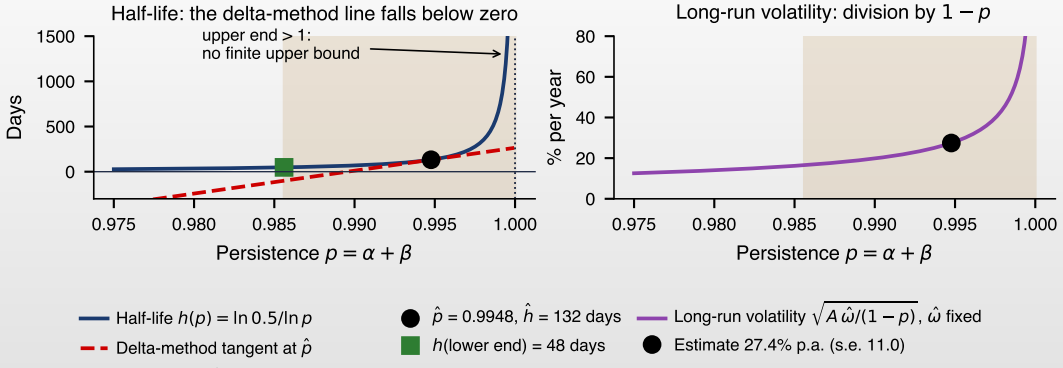
S&P 500: GARCH(1,1)-t Estimates

Parameter	Estimate	Robust SE	Classical SE	Ratio
ω	0.0157			
α	0.122	0.0107	0.0101	1.06
β	0.872	0.0103	0.0096	1.08
ν	6.25	0.48		
$\alpha + \beta$	0.9948			

- Mean $\hat{\mu} = 0.079\%$ per day
  - Full table with every standard error: Seminar 5, B1
- Large $\hat{\beta}$: the variance memory dominates
  - $\hat{\alpha}$: a moderate news reaction
  - $\hat{\nu} = 6.25$: tails much heavier than Normal
- Half-life 132 days
  - Long-run volatility $\sqrt{251.5\,\hat{\omega}/(1-\hat{\alpha}-\hat{\beta})} = 27.4\%$ p.a. ($A = 251.5$ returns a year)
  - Above the sample 19.2%: dividing by $1-\hat{\alpha}-\hat{\beta}$ makes it imprecise (next slide)

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Near-Unit Persistence: What the Delta Method Can and Cannot Do (1/2)



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Near-Unit Persistence: What the Delta Method Can and Cannot Do (2/2)

- $g(\theta)$: a smooth function of the parameters, with gradient ∇g
  - E.g. the long-run volatility or the half-life
- Delta method: $\widehat{\mathrm{SE}}\, g(\hat{\theta}) \approx \sqrt{\nabla g' \hat{V} \nabla g}$, with $\hat{V}$ the robust covariance of $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$
- For the half-life: $p = \alpha + \beta$ and $h = \ln 0.5 / \ln p$
- Long-run volatility 27.4% p.a., SE 11.0 percentage points
- Half-life: $\partial h / \partial p = 25\,330$ at $\hat{p} = 0.994769$
  - $\widehat{\mathrm{SE}}(\hat{p}) = 0.0047$, so $\widehat{\mathrm{SE}}(\hat{h}) \approx 118$ days
  - $\hat{h} \pm 1.96\, \widehat{\mathrm{SE}}$ includes negative half-lives: the linearisation fails
- Transform the interval for p instead: $[0.9856; 1.0039]$ gives $h \in [48; \infty)$
  - Whether $p = 1$ is plausible needs a boundary test (next slides)

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Testing on the Boundary (1/2): a Decision Table

Situation under H_0	**Example**	**Likelihood Ratio (LR):** $LR = 2(\ell_1 - \ell_0)$	**5% critical value**
Interior restriction	GJR (asymmetric GARCH) vs GARCH, $\gamma = 0$ (with $\alpha > 0$)	χ^2_1	3.84
One parameter on the boundary, the others identified and interior	Normal vs Student-t ($1/\nu = 0$); IGARCH ($\alpha + \beta = 1$)	$\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ Self & Liang (1987), Andrews (2001)	2.71
A nuisance parameter unidentified	No ARCH: $\alpha = 0$ leaves β unidentified	non-standard Andrews (2001)	simulate
Several bounds active at once	GJR with $\alpha_0 = 0$ under H_0, test of $\gamma = 0$	non-standard	simulate

- ℓ_1, ℓ_0: maximised log-likelihoods without and with the restriction
  - $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$: $LR = 0$ with probability $\frac{1}{2}$, otherwise χ^2_1
- GJR: γ is the extra variance reaction to negative shocks (Asymmetry section)
- The LR law depends on the true parameter under H_0, not on the estimate of the alternative: on the S&P 500, GARCH-t fitted under $\gamma = 0$ has $\hat{\alpha} = 0.122 > 0$
  - $\hat{\alpha} = 0$ in GJR affects the Wald t for α, not automatically the LR law for $\gamma = 0$

Testing on the Boundary (2/2): Why IGARCH Is a Boundary Case

- IGARCH is on the boundary only because the fit restricts $\alpha + \beta \leq 1$, as `arch` does
  - Under $H_0 : \alpha + \beta = 1$ the unrestricted estimate would exceed 1 about half of the time; the restriction sets it to 1 and then $LR = 0$: hence the mixture and 2.71
  - Without that restriction, $\alpha + \beta = 1$ lies inside the strictly stationary region, the QMLE is still asymptotically Normal there Lumsdaine (1996), and the LR would be compared with χ^2_1 (3.84)
  - QMLE normality is not enough for a quasi-LR: the χ^2_1 law requires a correctly specified density or a corrected statistic
- A Wald t for an estimate at the bound (e.g. $\hat{\alpha} = 0$) is not $N(0, 1)$
  - A parametric bootstrap from $\hat{\theta}$ is inconsistent when θ_0 is on the boundary Andrews (2000)
  - Simulate under H_0 instead

Worked Example (1/2): The One-Day Forecast

- S&P 500, GARCH(1,1)-t: $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.122$, $\hat{\beta} = 0.872$, $\hat{\mu} = 0.079$
  - Persistence $\hat{p} = \hat{\alpha} + \hat{\beta} = 0.994769$
  - With the rounded values shown, $1 - \hat{p} = 0.006$ and the long-run variance would be 2.62 instead of 3.00
- Last day $T = 18.09.2026$: $\hat{\varepsilon}_T = r_T - \hat{\mu} = 0.088$, filtered variance $\hat{\sigma}^2_T = 0.565$
  - Filtered variance: the recursion run through the whole sample with $\hat{\theta}$, read off on the last day T
  - $\hat{\sigma}^2_{T+1} = \hat{\omega} + \hat{\alpha}\hat{\varepsilon}^2_T + \hat{\beta}\hat{\sigma}^2_T = \hat{\omega} + 0.0009 + 0.493 = 0.509$, i.e. 11.3% p.a.
- A quiet last day: $\hat{\sigma}^2_{T+1}$ is far below the long-run level, so the forecasts will rise (next slide)

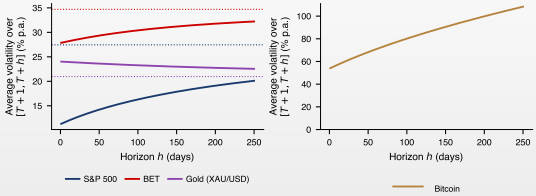
MFM_ch5_garch_estimation

Worked Example (2/2): Ten Days Ahead

- Long-run variance $\hat{\sigma}^2 = \hat{\omega}/(1 - \hat{\alpha} - \hat{\beta}) = 3.00$
  - Day 10: $\hat{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^9(\hat{\sigma}^2_{T+1} - \hat{\sigma}^2)$
  - $= 3.00 + 0.954 \times (0.509 - 3.00) = 0.624$
- 10-day variance $\sum_{h=1}^{10} \hat{\sigma}^2_{T+h} = 5.67$, volatility 2.38%
  - The sum in closed form (toy check in the persistence section): $10\,\hat{\sigma}^2 + (\hat{\sigma}^2_{T+1} - \hat{\sigma}^2)\,(1 - \hat{\rho}^{10})/(1 - \hat{\rho})$
  - $= 29.95 + (0.509 - 2.995) \times 9.768 = 5.67$
  - The square-root-of-time rule gives $\sqrt{10} \times 0.714 = 2.26\%$: too low, because $\hat{\sigma}^2_{T+1} < \hat{\sigma}^2$ and volatility is forecast to rise

MFM_ch5_garch_estimation

Volatility Term Structure (1/2)



- Average volatility over the next h days, forecast from 18.09.2026 (GARCH(1,1)-t)
  - Dotted: long-run level
- S&P 500: from 11.3% for the next day to 20.1% on average over 250 days
  - Below its long run today
- Gold falls from 24.0% to 22.5%

MFM_ch5_persistence ▶ Interactive

Volatility Term Structure (2/2)

- Bitcoin and EUR/RON: the estimate hits the bound $\hat{\alpha} + \hat{\beta} = 1$ (IGARCH)
  - With $\hat{\omega} > 0$ (for EUR/RON, $\hat{\omega} \approx 5 \cdot 10^{-8}$ in $\%^2$)
  - So the variance forecast grows linearly by $\hat{\omega}$ per day
- Bitcoin: 54.1% for the next day, 108.3% on average over 250 days (about 8 calendar months)
  - EUR/RON is not in the chart
- This unbounded growth is not a usable forecast
  - In Chapter 8, GARCH forecasts are withheld on days when the fit is not admissible
  - Bitcoin: 1 660 of 3 384 evaluation days
  - EUR/RON: 20 of 4 352

MFM_ch5_persistence

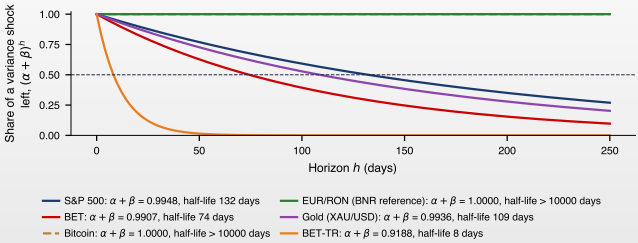
GARCH(1,1)-t in Six Markets

Series	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	$h_{1/2}$ (days)	$\hat{\nu}$	Long-run vol.
S&P 500	0.0157	0.122	0.872	0.9948	132	6.25	27.4
BET	0.0448	0.192	0.799	0.9907	74	5.01	34.7
BET-TR	0.0755	0.206	0.713	0.9188	8	5.02	15.2
Bitcoin	0.1937	0.109	0.891	1.0000	∞	3.18	–
EUR/RON	0.0000	0.123	0.877	1.0000	∞	4.20	–
Gold	0.0107	0.050	0.944	0.9936	109	4.52	21.0

- Long-run volatility in % p.a., each series annualised with its own frequency
  - –: when $\hat{\alpha}+\hat{\beta} \geq 0.999$
  - EUR/RON: $\hat{\omega}$ in $\%^2$, after the fit on $10\, r_t$
- S&P 500, BET and gold: persistence 0.99–0.995
  - The BET reacts more to news ($\hat{\alpha} = 0.192$) and forgets faster than the S&P 500
- BET-TR covers only 2014–2026: a comparison with the BET needs the same sample
- EUR/RON: on 6 May 2025 a move of $+1.20\%$ after days of $\pm 0.004\%$ gives a standardised residual of 154
- What the result supports: description

MFM_ch5_garch_estimation

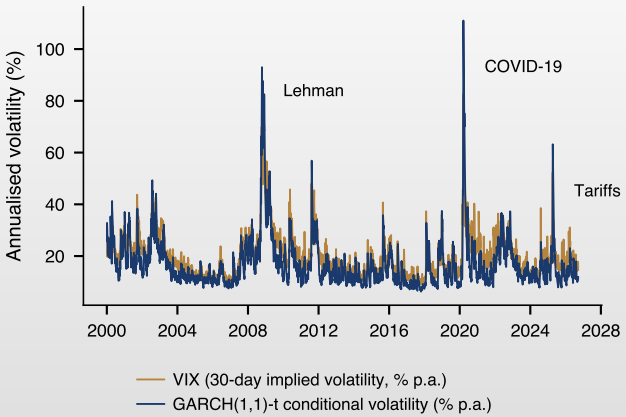
Persistence: How Fast Do Shocks Fade?



- $(\hat{\alpha}+\hat{\beta})^h$: the share of a variance shock left after h days
  - Dashed line: one half
- Half-lives: column $h_{1/2}$ of the six-market table
  - In Bitcoin and EUR/RON shocks never fade in the model
- Level shifts in volatility can mimic high persistence Lamoureux & Lastrapes (1990)
  - Regime switching: Hamilton & Susmel (1994)
  - Report half-lives as intervals (Seminar 5, B4)

MFM_ch5_persistence ▶ Interactive

S&P 500: GARCH Volatility and the VIX (1/2)



- $\hat{\sigma}_t\sqrt{A}$: annualised GARCH(1,1)-t volatility
- VIX: 30-day implied volatility (Volatility Index of the Chicago Board Options Exchange)
- Chart labels
  - Lehman: Lehman Brothers bankruptcy, September 2008
  - COVID-19: Coronavirus Disease 2019, March 2020
  - Tariffs: US tariff announcement, April 2025

MFM_ch5_sp500_vix ▶ Interactive

S&P 500: GARCH Volatility and the VIX (2/2)

- Correlation of the two series, 2000–2026: 0.89
- $\hat{\sigma}_t$ is filtered with parameters estimated on the full sample (in-sample), not an out-of-sample forecast
- The VIX is higher on average: 19.8% against 17.0%
  - The VIX exceeds GARCH volatility on 84% of the days
- Not a measured variance risk premium
  - The premium compares VIX^2 with a 30-day variance forecast under the real-world probability
  - $\hat{\sigma}_t$ is a one-day volatility
- What the result supports: association

MFM_ch5_sp500_vix

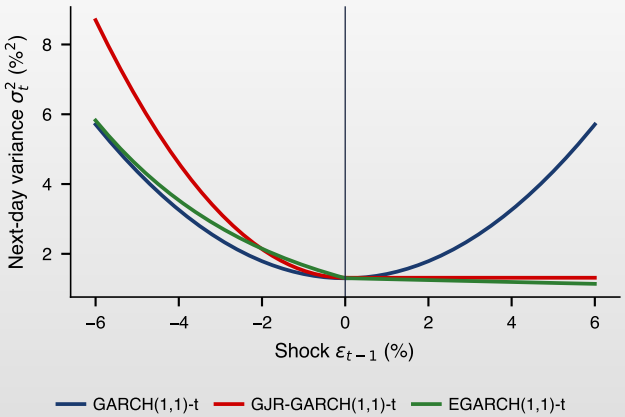
GJR-GARCH (Glosten, Jagannathan and Runkle, 1993)

- **The question**: does volatility rise more after a fall than after a rise?
  - Shocks of the same size (stylised fact 9, Chapter 1)
- GARCH(1,1) cannot see the sign: ε_{t-1}^2 is the same for -3 and $+3$
- GJR: a negative shock gets the extra slope γ Glosten, Jagannathan & Runkle (1993)
  - $\sigma_t^2 = \omega + (\alpha + \gamma\, I_{t-1})\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2$, with $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$, else 0
  - Good news: slope α; bad news: slope $\alpha + \gamma$; positivity: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$ (γ may be negative)
  - Persistence $\alpha + \gamma/2 + \beta$ for symmetric z_t; for skewed z_t replace $1/2$ by $\mathbb{E}[z_t^2 1(z_t < 0)]$, $1(\cdot)$ the indicator (Seminar 5, A3)
- S&P 500: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ (t = 10.0), $\hat{\beta} = 0.881$
  - Symmetry test: H_0: $\gamma = 0$ by t or by LR against GARCH(1,1)
- $\hat{\alpha} = 0$ is on the boundary: the t-test of γ is not standard
  - The χ_1^2 law of the LR for $\gamma = 0$ assumes $\alpha_0 > 0$
  - See the decision table for tests on the boundary
- Candidate mechanisms, not tested here
  - Leverage: a falling price raises the debt-to-equity ratio
  - Volatility feedback: higher expected volatility raises required returns, so prices fall now

EGARCH (Nelson, 1991)

- Exponential GARCH (EGARCH) models the log of the variance Nelson (1991)
  - Driven by the size and the sign of yesterday's standardised shock
  - $\ln \sigma_t^2 = \omega + \alpha\left(|z_{t-1}| - \mathbb{E}|z_{t-1}|\right) + \gamma\, z_{t-1} + \beta \ln \sigma_{t-1}^2$
  - $z_{t-1} = \varepsilon_{t-1}/\sigma_{t-1}$: standardised shock; $|z_{t-1}| - \mathbb{E}|z_{t-1}|$: its size above the average size
  - Size effect α, sign effect γ; $\gamma < 0$: negative shocks raise volatility more
- The log form keeps $\sigma_t^2 > 0$ without any sign restriction on the parameters
- Persistence: β; $|\beta| < 1$ gives stationarity of the log-variance $\ln \sigma_t^2$
- Moments of returns need separate conditions: with Student-t innovations the unconditional variance can be infinite even if $\nu > 4$
- S&P 500: $\hat{\alpha} = 0.138$, $\hat{\gamma} = -0.165$ (t $= -15.2$), $\hat{\beta} = 0.979$
- In `arch`, $\mathbb{E}|z|$ is set to $\sqrt{2/\pi}$, its value under the Normal distribution

The News Impact Curve for the S&P 500



- **News impact curve** Engle & Ng (1993): σ_t^2 as a function of ε_{t-1}
  - σ_{t-1}^2 fixed at the sample variance $s^2 = 1.47$ (%^{2})
  - Not at the long-run variance of the model
- Shock of -3%: GJR 3.16, EGARCH 2.75
  - $+3$%: GJR 1.31, EGARCH 1.22
  - GARCH: 2.40 for both
- After bad news the forecast is 2.4 times (GJR) and 2.3 times (EGARCH) that after good news
  - Ratios of levels, not of increases

MFM_ch5_asymmetry ▶ Interactive

Asymmetry in Six Markets

Series	$\hat{\gamma}_{GJR}$	t	LR vs GARCH	p-value	$\hat{\gamma}_{EGARCH}$	t
S&P 500	0.205	10.0	236.2	< 0.001	-0.165	-15.2
BET	0.043	2.1	4.9	0.026	-0.028	-2.6
BET-TR	0.111	2.9	10.3	0.001	-0.063	-3.5
Bitcoin	-0.013	-0.7	0.7	0.416	0.015	1.2
EUR/RON	0.039	3.0	9.3	0.002	-0.040	-2.5
Gold	-0.043	-4.7	26.7	< 0.001	0.038	5.7

- Student-t innovations
  - Robust t
  - $LR \sim \chi^2_1$ (3.84) if the Student-t density is correct
- The robust t is $N(0,1)$ only with the other parameters interior (not for the S&P 500 GJR, where $\hat{\alpha} = 0$)
- Moments: Gaussian QMLE requires $\mathbb{E}z_t^4 < \infty$; a correctly specified Student-t MLE does not; with a wrong density, $\hat{\nu} > 4$ alone does not guarantee the robust errors Fan, Qi & Xiu (2014)
- Bad news weighs more (GJR $\hat{\gamma} > 0$, EGARCH $\hat{\gamma} < 0$)
  - S&P 500 strongly
  - Smaller but significant at 5% for the BET, BET-TR and EUR/RON
  - Bitcoin: none detected
  - Gold: inverted sign
- What the result supports: association
  - Leverage for equities and a safe-haven role for gold: interpretations consistent with these signs, not tested mechanisms

MFM_ch5_asymmetry

S&P 500: Twelve Models Compared by BIC

ΔBIC	$N(0,1)$	Student-t	Skewed-t	GED
GARCH(1,1)	650.2	347.1	307.9	339.6
GJR-GARCH(1,1)	395.3	119.8	49.6	140.8
EGARCH(1,1)	373.4	74.7	0.0	110.7

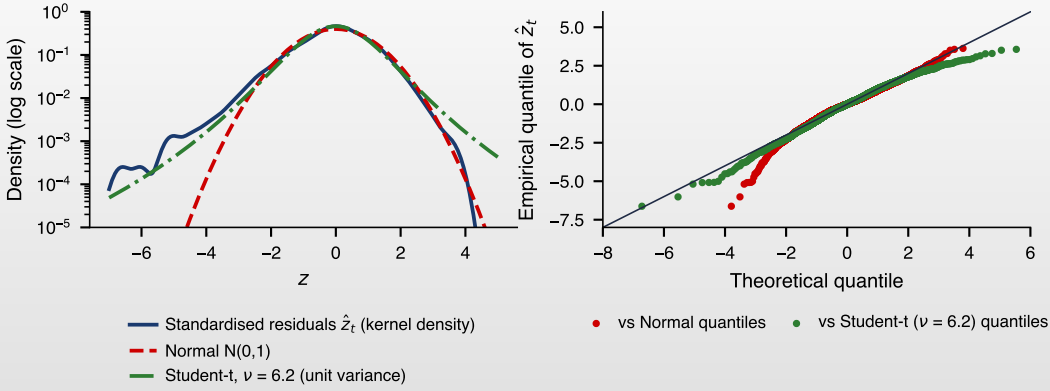
- ΔBIC: Bayesian Information Criterion minus the lowest BIC among the 12 models
  - BIC $= -2\ell + k \ln n$ (k parameters)
  - 0 = preferred model
- Akaike Information Criterion, AIC $= -2\ell + 2k$: we choose by BIC
  - With $n = 6\,717$ ($\ln n \approx 8.8$), BIC penalises each parameter 4.4 times more
  - It avoids over-parameterised models
- Two choices matter
  - Student-t instead of Normal innovations lowers BIC by 276–303
  - Asymmetry lowers it by a further 227.4–272.4 (Student-t column, GJR and EGARCH against GARCH)
  - Winner: EGARCH with skewed-t innovations
- BIC ranks in-sample fit, not forecast accuracy
  - The forecast section tests whether these gains survive out-of-sample
- What the result supports: comparison of procedures
- Seminar 5, B3 re-estimates the BET row of the asymmetry table and explains the LR, AIC and BIC decisions

MFM_ch5_garch_estimation

Standardised Residuals: What We Check

- **The question**: has the chosen model removed all dynamics from $\hat{z}_t$, and does $\hat{z}_t$ have the assumed density?
- **Standardised residual**: $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$
  - If the model is right, $\hat{z}_t$ is close to i.i.d. with variance 1
- Checks, each with H_0: the model captures that feature
  - Mean equation: Ljung–Box $Q(10)$ on $\hat{z}_t$; variance equation: $Q(10)$ on $\hat{z}_t^2$ and LM(5); distribution: Quantile–Quantile (QQ) plot against the assumed density
- Sign-bias regression Engle & Ng (1993): $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + b_3 S_{t-1}^+ \hat{\varepsilon}_{t-1} + u_t$
  - $S_{t-1}^- = 1$ if $\hat{\varepsilon}_{t-1} < 0$, else 0; $S_{t-1}^+ = 1 - S_{t-1}^-$; u_t: regression error; the size terms use the unstandardised residual $\hat{\varepsilon}_{t-1}$ (eq. 18 in Engle & Ng (1993))
  - b_1 (sign bias): $\hat{z}_t^2$ larger on average after a negative shock, and the sign-bias test is the t of b_1; b_2, b_3: negative and positive size bias; joint test $nR^2 \sim \chi_3^2$ (7.81 at 5%), conservative because it omits the gradient of the fitted variance

S&P 500: Are the Innovations Normal? (1/2)



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S&P 500: Are the Innovations Normal? (2/2)

- Standardised residuals of GARCH(1,1) with Normal innovations
  - Excess kurtosis falls from 10.7 (returns) to 1.73
  - Skewness -0.53, Jarque–Bera 1 156
- Jarque–Bera $= n[S^2/6 + E^2/24]$, S: skewness, E: excess kurtosis
  - χ^2_2 under Normality, 5% critical value 5.99
  - Normality is rejected by a wide margin
- 16 residuals below -4
  - Expected: 0.21 under the Normal, 8.6 under a Student-t with $\nu = 6.2$
  - The smallest standardised residual: $\hat{z} = -6.6$, on 27 February 2007
- A longer left tail: skewed-t with $\hat{\lambda} = -0.155$
  - EGARCH, $\hat{\nu} = 7.5$ degrees of freedom
  - Hence BIC prefers skewed-t innovations

MFM_ch5_innovations

S&P 500: Diagnostics of Five Models

Model	$Q(10)$ on $\hat{z}$	$Q(10)$ on $\hat{z}^2$	LM(5)	Sign bias t	Joint χ^2_3
GARCH-N	21.1	16.2	8.1	3.65	41.6
GARCH-t	21.8	13.5	5.8	3.57	40.6
GJR-t	22.0	10.1	4.4	3.61	28.1
EGARCH-t	27.4	10.5	8.3	2.05	9.9
GJR-skewed-t	19.4	10.6	5.0	3.90	31.9

- 5% critical values: 18.3 (χ^2_{10}), 11.1 (χ^2_5), 7.81 (χ^2_3), $|t| > 1.96$
  - The χ^2 critical values ignore parameter estimation, so the tests are conservative (Further Topics)
- Mean: $Q(10)$ on $\hat{z}$ between 19.4 and 27.4 in every model
  - Above the critical value 18.3: a little autocorrelation that a constant mean ignores
- Variance: $Q(10)$ on $\hat{z}^2$ falls from 6 135 (on r_t^2) to 10–16
  - LM(5) is not significant
  - These naive tests detect no clustering left
- Asymmetry: the joint test rejects all five models at 5%
  - GARCH-t 40.6, GJR-t 28.1, EGARCH-t 9.9 (p-value 0.019, the smallest statistic)
- The sign-bias test (t of b_1) stays above 1.96 in every model (2.05–3.90)
  - With $n = 6\,717$, judge economic size, not only p-values

MFM_ch5_innovations

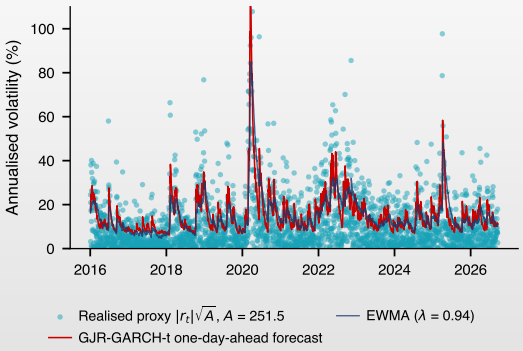
The Proxy Problem and Robust Losses

- **The question**: do the BIC gains of GJR and EGARCH survive out-of-sample?
- Volatility is never observed: we compare forecasts with a noisy **proxy**
  - Here r_t^2, since we use no intraday data
  - Realised variance: Chapter 9
- $r_t^2 = \sigma_t^2 z_t^2$ is unbiased ($\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$ for $\mu = 0$) but very noisy
  - Even the perfect forecast $h_t = \sigma_t^2$ explains little of the variation of r_t^2 (a small R^2), because z_t^2 is noisy Andersen & Bollerslev (1998)
  - S&P 500: we ignore the mean, since $\hat{\mu}^2 = 0.0062$ against an average daily variance of 1.47 ($\%^2$), a bias below 1%
- Patton (2011): only some loss functions rank forecasts correctly with a noisy proxy
  - Robust: Mean Squared Error (MSE) $(r_t^2 - h_t)^2$ and **Quasi-Likelihood (QLIKE)** $r_t^2/h_t + \ln h_t$, with h_t the variance forecast for day t
  - Not robust: $(|r_t| - \sqrt{h_t})^2$ and $(\ln r_t^2 - \ln h_t)^2$: their ranking depends on the proxy noise
- QLIKE is (up to constants) minus the Normal log-likelihood
  - Its expected value is lowest at $h_t = \sigma_t^2$
  - It penalises under-prediction more

Forecast Evaluation: Mincer–Zarnowitz and Diebold–Mariano

- **Mincer–Zarnowitz (MZ) regression** Mincer & Zarnowitz (1969): regress the proxy on the forecast
  - $r_t^2 = a + b\, h_t + u_t$
- Unbiased forecast: $a = 0$, $b = 1$
  - Joint Wald test with Heteroskedasticity and Autocorrelation Consistent (HAC) errors; χ^2_2 under H_0
  - $b < 1$: forecasts too dispersed (too high when high, too low when low)
- **Diebold–Mariano test** Diebold & Mariano (1995): do two forecasts have the same expected loss?
  - Loss differential $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$
  - L: loss function (QLIKE or MSE); h_t^A, h_t^B: forecasts of models A and B; $d_t < 0$: A did better on day t
- Diebold–Mariano (DM) statistic: $DM = \bar{d} / \widehat{\mathrm{SE}}(\bar{d})$
  - $\bar{d}$: mean of d_t over the forecast days; HAC standard error, since d_t is autocorrelated
  - $\approx N(0,1)$ under equal accuracy; $DM < -1.96$: A significantly better; $DM > 1.96$: B significantly better
- The $N(0,1)$ law assumes given forecasts
  - For estimated, nested models (GARCH-t inside GJR-t) it can fail
  - So DM t is read descriptively there (Giacomini–White, GW: Further Topics)

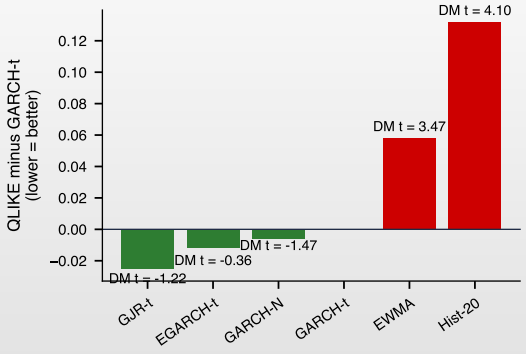
Out-of-Sample Design and Forecast Paths, 2016–2026



- S&P 500: 2 693 one-day forecasts
  - Refit each January on all data since 2000 (expanding window)
  - Daily updates with fixed parameters
- Timing: h_t uses data up to the close of day $t-1$ and is compared with r_t^2
  - Changing r_t must leave h_t unchanged
- Models: GARCH-N, GARCH-t, GJR-t, EGARCH-t
- Benchmarks without estimated parameters: EWMA ($\lambda = 0.94$) and Hist-20, the mean of r^2 over the last 20 days

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S&P 500: Does Anything Beat GARCH(1,1)? (1/2)



- Mean QLIKE minus that of GARCH-t, with the DM t (HAC, 5 lags)
  - Lowest QLIKE: GJR-t (0.731)
- GARCH family: DM t from -1.47 to -0.36, descriptive for the nested pairs (GARCH-N and GJR-t against GARCH-t)
  - EGARCH-t is not nested in GARCH-t
  - The DM test requires finite moments and limited dependence of $d_t = r_t^2(1/h_t^A - 1/h_t^B) + \ln(h_t^A/h_t^B)$
  - $\mathbb{E}r_t^4 < \infty$ is not the relevant condition: the forecasts normalise r_t^2; HAC does not repair moments that do not exist
- EWMA and Hist-20 are clearly worse ($t = 3.47$, 4.10)
  - Mean reversion and estimated parameters pay off Hansen & Lunde (2005)
- What the result supports: comparison of procedures

MFM_ch5_forecast_evaluation ▶ Interactive

S&P 500: Does Anything Beat GARCH(1,1)? (2/2)

- With MSE the ranking changes: EGARCH-t 21.30, GJR-t 21.80, GARCH-N 22.27, GARCH-t 22.76 ($\%^4$)
- MZ: unbiasedness not rejected for GARCH-type forecasts, rejected for Hist-20
  - All statistics: Appendix
- Not rejecting is weak evidence: SE of b is 0.25 for GARCH-t
  - MZ inference requires finite moments of the score $(1, h_t)' u_t$, $u_t = r_t^2 - a - b\, h_t$; even for the ideal forecast, the slope term involves $\mathbb{E} h_t^4$
  - $\mathbb{E} r_t^4 < \infty$ is not enough in general, and HAC does not repair moments that do not exist (heavy tails: Chapter 1)
  - A few crisis days drive b (Seminar 5, B5)

MFM_ch5_forecast_evaluation

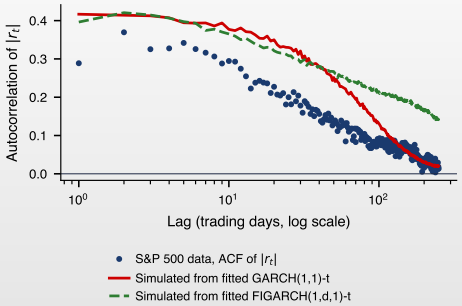
Further Topics: GW, MCS, monthly forecasts

GARCH-in-Mean: Is Risk Rewarded?

- **The question**: what does GARCH(1,1) miss, and which extension answers each limit?
- **GARCH-in-Mean (GARCH-M)** Engle, Lilien & Robins (1987): $r_t = c + \kappa\,\sigma_t + \varepsilon_t$ (or $\kappa\,\sigma_t^2$)
  - c: constant; κ: price of risk, the extra expected return per unit of volatility
  - $\kappa > 0$: investors demand a higher expected return when risk is higher (risk-return trade-off)
- S&P 500, GARCH(1,1)-t in mean: $\hat{\kappa} = 0.096$ (t = 3.03) with σ_t
  - 0.028 (t = 2.36) with σ_t^2
  - Positive and significant in this sample, but the literature is mixed: Glosten, Jagannathan & Runkle (1993) found a negative relation
- A daily mean effect is small and fragile
  - It is sensitive to the sample and to the asymmetry in variance
- What the result supports: association

MFM_ch5_long_memory

Long Memory: the Shape of the ACF of $|r_t|$



- S&P 500, ACF of $|r_t|$ (log scale): in the data, 0.289 at lag 1, 0.083 at lag 100 and 0.013 at lag 250
- A simulated GARCH(1,1)-t overstates the ACF at every lag
  - 0.417 at lag 1, 0.131 at lag 100, 0.020 at lag 250
- The problem is the shape: a single $\alpha + \beta$ cannot give both a low level at lag 1 and a slow decay
- **Fractionally Integrated GARCH (FIGARCH)** Baillie, Bollerslev & Mikkelsen (1996)
  - The filter $(1 - L)^d$, with $Lx_t = x_{t-1}$ and $d \in (0, 1)$, gives a hyperbolic decay
- $\hat{d} = 0.60$ (SE 0.06), BIC lower by 34.9
- FIGARCH overstates the memory even more: 0.214 at lag 100, 0.144 at lag 250
- No single-scale model fits all horizons
  - Realised volatility and Heterogeneous AutoRegressive (HAR) models in Chapter 9

MFM_ch5_long_memory ▶ Interactive

Limits of GARCH and the Extensions They Motivate

Limit	Evidence in this chapter	Extension
Level shifts inflate $\alpha + \beta$ Lamoureux & Lastrapes (1990)	BET-TR (2014–2026) far less persistent than the BET (2000–2026): a hint, not proof, as sample and index differ	GARCH-MIDAS (Mixed Data Sampling), a slow long-run component Engle, Ghysels & Sohn (2013); regime switching Hamilton & Susmel (1994)
Reaction lag of one day	COVID-19: the VIX moved the same day, GARCH the next	Realized GARCH, with a realised-variance measurement equation Hansen, Huang & Shek (2012) (Chapter 9)
One outlier drives the update	EUR/RON residual of 154 (May 2025)	GAS (Generalised Autoregressive Score) models, bounded score Creal, Koopman & Lucas (2013) (case study below)
Shape of the ACF of $\lvert r_t \rvert$	GARCH-t and FIGARCH overstate the ACF (previous slide)	HAR on realised variance (Chapter 9)
One asset at a time	portfolios need covariances	Multivariate GARCH and DCC, Dynamic Conditional Correlation (Chapter 6)
Managed prices	EUR/RON: long calm, then jumps	Model interventions as jumps or regimes

- Each extension is judged like GARCH: out-of-sample, robust loss, Model Confidence Set
  - Model equations: Appendix

MFM_ch5_garch_estimation

Three Research Extensions and What They Assume

- **Quasi score-driven volatility**: a replication of a 2023 journal article
  - Builds on: the Student-t density, LR tests and their boundary versions, the score-driven (GAS) update of the extensions section
  - Continued in Seminar 5, C3
- **Volatility cycles (MF2-GARCH)**: a slow component learnt from the forecast errors of GARCH, a replication of a 2025 journal article
  - MF2-GARCH: Multiplicative Factor Multi-Frequency GARCH
  - Builds on: GJR-GARCH, QMLE, the BIC, QLIKE and the Diebold–Mariano test
- **AI for scientific discovery**: an assistant-driven research loop with checks
  - Builds on: the out-of-sample design, QLIKE, Diebold–Mariano and the Model Confidence Set
  - Its checklist supports the AI audit of Seminar 5, C1 and the team project

Score-Driven Volatility: Should Shocks Be Downweighted?

- **Question** Blasques, Francq & Laurent (2023)
  - Must a score-driven volatility model update σ_t^2 with the score of the same density that describes the returns?
- Score-driven (GAS) update Creal, Koopman & Lucas (2013)
  - Tomorrow's parameter moves in the direction that would have made today's observation more likely
  - $f_{t+1} = \omega + \alpha\, S(f_t)\, \partial \log p(y_t \mid f_t)/\partial f_t + \beta f_t$
  - y_t: the return; f_t: the time-varying parameter (here the variance); $p(y_t \mid f_t)$: the density of y_t given f_t
  - $\partial \log p/\partial f_t$: the score; $S(f_t) > 0$: a scaling factor; ω, α, β: as in GARCH
- With a Student-t density, the same parameter ν sets both
  - The tail thickness of z_t
  - How strongly a large shock is downweighted in the update

QSD: Separating Tail Thickness from Shock Downweighting

- **Quasi Score-Driven (QSD) GARCH(1,1)-t**, eq. (30) of the paper
  - $y_t = \sqrt{f_t}\,\epsilon_t$, ϵ_t standardised Student-t with $1/\xi$ degrees of freedom
  - f_t is σ_t^2, ϵ_t is z_t
  - $f_{t+1} = \omega + \varpi X_t + \alpha \dfrac{1+\zeta}{1-2\zeta+\zeta\epsilon_t^2}\epsilon_t^2 f_t + \beta f_t$
  - X_t: an exogenous variable, the squared VIX at a daily horizon; ϖ: its coefficient
  - ξ: the tail thickness of ϵ_t ($\xi = 0$: Normal); ζ: the downweighting of large ϵ_t^2 in the update
  - $\zeta = 0$: GARCH(1,1)-t; $\zeta = \xi$: Beta-t-GARCH (score-driven)
  - Why: with $\zeta = \xi = 1/\nu$, multiplying by ν gives $\dfrac{1+\zeta}{1-2\zeta+\zeta\epsilon_t^2} = \dfrac{\nu+1}{\nu-2+\epsilon_t^2}$, the weight in the Student-t score
- Data: 400 S&P 500 stocks, 3 January 1995 (or later) – 28 February 2019
  - Maximum likelihood with an AR(1) mean
  - LR tests of $\zeta = 0$ and $\xi = \zeta$ against χ_1^2 (full method: ▸ Appendix)
  - Probability Integral Transform (PIT) test: $u_t = F(y_t \mid \mathcal{F}_{t-1})$, the fitted distribution function at the observed return, must be i.i.d. uniform on $(0,1)$ if the density is right

QSD: Key Results (Paper, Tables 4 and 6)

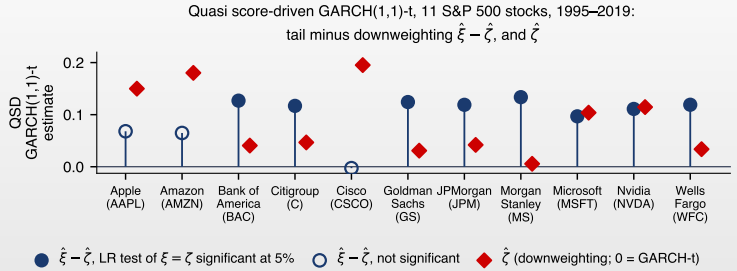
Rejected at 5%, % of the 400 stocks	Student-t (Table 4)	Skewed t (Table 6)
LR, H_0: $\zeta = 0$ (GARCH update)	89.5	98.8
LR, H_0: $\xi = \zeta$ (score-driven; skewed: also $\kappa = \tau$)	53.5	100.0
PIT test: GARCH(1,1)	40.0	33.1
PIT test: Beta-t-GARCH	46.0	72.2
PIT test: QSD GARCH(1,1)	45.7	11.3

- κ: asymmetry of the density
  - τ: asymmetry of the update
  - $\kappa = \tau = 0.5$: symmetric (skewed-t version, Table 6)
- Downweighting large shocks matters (Table 4)
  - $\zeta = 0$ is rejected for most stocks (first row)
  - Every $\hat{\zeta}$ is positive
- What the result supports: comparison of procedures
  - Beating GARCH-t and Beta-t-GARCH does not make QSD adequate: the PIT test rejects it for 45.7% of the stocks (Table 4)

QSD: Reading the Results (Paper, Figs. 2–6)

- **Beta-t-GARCH downweights too much** (Figs. 2–3)
  - Where $\hat{\xi} - \hat{\zeta}$ is significant, $\hat{\xi} > \hat{\zeta}$ for all stocks but one
  - Median $1/\hat{\zeta}$: 9.48 in QSD, against 5.07 in Beta-t-GARCH, where $\zeta = \xi$
  - Tail thickness and shock downweighting are two different parameters
- **Skewed t** (Figs. 5–6)
  - $\hat{\tau}$ far below 0.5 (median 0.333), while $\hat{\kappa}$ stays near 0.5
  - Leverage enters through the update equation, not through the density
- **Calibration** (Table 6)
  - With skewed-t updates, the PIT test rejects the QSD model for only 11.3% of the stocks, against 72.2% for Beta-t-GARCH

Case Study: Replication on 11 Stocks (cf. Paper, Fig. 2)



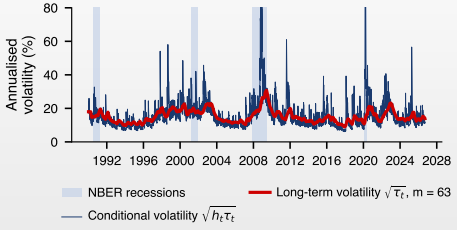
- Circles: $\hat{\xi} - \hat{\zeta}$ (tails minus downweighting) for each stock
  - Filled if the LR test rejects $\xi = \zeta$ at 5%
- Diamonds: $\hat{\zeta}$, the downweighting of large shocks ($\zeta = 0$: GARCH-t)
- $\hat{\xi} > \hat{\zeta}$ for 10 of 11 stocks
  - The exception is Cisco ($\hat{\xi} = 0.193$, $\hat{\zeta} = 0.195$)

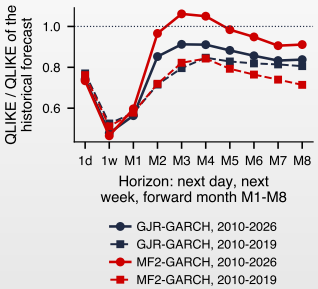
MFM_ch5_quasi_score_driven ▶ Interactive

Volatility Cycles: Can GARCH Learn from Its Own Errors?

- Conrad & Engle (2025): the standardised forecast errors of a one-component GARCH, $r_t^2/\tilde{h}_t$, are predictable
  - the predictor is their average over the last m days: the model underestimates volatility for months in recessions and overestimates it in expansions
- Why it matters: risk forecasts several months ahead depend on the level to which volatility reverts
  - GJR-GARCH always reverts to the same unconditional variance, whatever the phase of the cycle
- MF2-GARCH (Multiplicative Factor Multi-Frequency GARCH): $r_t = \mu + e_t$, $e_t = \sqrt{h_t\,\tau_t}\,Z_t$, with Z_t i.i.d., mean 0, variance 1, symmetric density
  - short-term component: $h_t = (1-\phi) + (\alpha + \gamma\,1\{e_{t-1} < 0\})\,e_{t-1}^2/\tau_{t-1} + \beta h_{t-1}$, with $\phi = \alpha + \gamma/2 + \beta$ and mean 1
  - long-term component: $\tau_t = \lambda_0 + \lambda_1 V_{t-1}^{(m)} + \lambda_2 \tau_{t-1}$
  - $V_{t-1}^{(m)} = \frac{1}{m}\sum_{j=1}^{m} e_{t-j}^2/h_{t-j}$: the average standardised error of h over the last m days
- The paper's numbers, S&P 500, January 1971 to June 2023: the BIC chooses $m = 63$ days (one quarter)
  - persistence: 0.924 in the short term (ϕ) and 0.982 in the long term ($\lambda_1 + \lambda_2$), against 0.982 in GJR-GARCH
  - out of sample, 2010–2019: the RMSE (root mean squared error) for month 1 is 31.5% below that of the historical forecast (ratio 0.685; GJR-GARCH: 0.798)

S&P 500: The Slow Volatility Component and Forecasts Months Ahead





- Left: the conditional volatility $\sqrt{h_t \tau_t}$ and the long-term component $\sqrt{\tau_t}$ of MF2-GARCH with $m = 63$, annualised, 1990–2026
  - $\sqrt{\tau_t}$: maximum 30.8% (4 June 2009), minimum 8.9% (15 December 2017)
- Right: mean QLIKE divided by that of the historical forecast (the mean of r_t^2 over the last 10 years), for the next day, the next week and months 1–8

MFM_ch5_mf2_garch ▶ Interactive

MF2-GARCH: The Paper's Method and the Course Replication

- The paper: Gaussian QMLE, Bollerslev–Wooldridge standard errors, m chosen by the BIC for $m \leq 160$
  - forecasts s days ahead have a closed form (Theorem 2), with $\kappa = \mathrm{E}[Z_t^4]$ estimated in sample
  - out of sample, 2010–2023: rolling window, re-estimation every 21 trading days, m re-chosen by the BIC
  - target: realised variance from 5-minute returns plus the overnight return; RMSE and QLIKE losses, relative to the historical forecast
- The course replication: S&P 500, daily log returns, 3 January 1990 to 18 September 2026 (9,245 days)
  - the same specification (MF2-GARCH-rw-m) and the same estimator, on 36 years instead of 52
  - out of sample: 2010–2026, rolling window of 20 years, re-estimation every 21 days, $m = 63$ chosen once, by the BIC on 1990–2009
  - target: the sum of squared daily returns; the course data have no 5-minute returns before 2020
  - QLIKE $= y/f - \ln(y/f) - 1$, with y the target and f the forecast; the QLIKE ratio omits days with $y = 0$: 1 of 4,203 at the one-day horizon
  - Diebold–Mariano test of MF2 against GJR on the difference of $y/f + \ln f$, defined also for $y = 0$; Newey–West with as many lags as the target has days
- The 2010–2019 results compare with Table 4 of the paper
  - 2010–2026 includes the March 2020 shock; in the tests of Table 5, the paper gives zero weight to days with realised variance above the 99th percentile

MF2-GARCH: Forecast Gains by Horizon and Period

- In sample, the BIC again chooses $m = 63$; the log-likelihood rises by 57.1 over GJR-GARCH, the BIC falls by 96
  - fast component: $\hat{\phi} = 0.935$ (half-life 10 days); GJR-GARCH: 0.978 (32 days)
  - slow component: $\hat{\lambda}_{\mathbf{1}} + \hat{\lambda}_{\mathbf{2}} = 0.989$, but $\mathbb{E}\tau_t$ follows an AR(63) recursion; its dominant root, 0.9966, corresponds asymptotically to halving in about 204 days
  - a shock to a single daily error raises τ_t for 63 days; the effect then halves after another 137 days
  - $\hat{\lambda}_{\mathbf{1}} = 0.067$ (standard error 0.051), $\hat{\lambda}_{\mathbf{2}} = 0.922$ (0.059); in the paper: 0.112 (0.054) and 0.870 (0.062)
- 2010–2019, relative QLIKE for month 8: 0.714 for MF2-GARCH, 0.806 for GJR-GARCH (DM t -2.66)
  - for month 1: 0.578 and 0.576 (DM t 0.07); relative RMSE for month 1: 0.721 and 0.762
  - in the paper, relative QLIKE for month 8: 0.859 and 0.918
- 2010–2026: the QLIKE ranking reverses from month 1 to month 8 (month 8: 0.911 against 0.838, DM t 0.99)
  - the relative RMSE stays lower for MF2-GARCH for months 1–5 (month 1: 0.916 against 0.960, DM t -1.16)
- What the result supports: a forecast comparison; the paper's direction holds at long horizons on 2010–2019, not on 2010–2026
- The limits
  - a target built from daily returns is noisier than realised variance, so the DM tests have less power
  - the return variance is finite only if $\Gamma_m < 1$ (Theorem 1 of the paper), a condition on top of $\phi < 1$ and $\lambda_{\mathbf{1}} + \lambda_{\mathbf{2}} < 1$: $\hat{\Gamma}_{\mathbf{63}} = 0.883$ on the full sample, between 0.881 and 0.950 in the 201 rolling windows
  - $\hat{\alpha} = 0.000$, on the boundary (in the paper 0.003): the usual standard errors do not apply to this parameter Andrews (2001)
  - QMLE and losses in variance units require a finite fourth moment, which Chapter 1 questions; $\hat{\kappa} = 5.03$

Open Questions and the Discovery Loop (1/2)

- Questions
  - do machine-learning or realised-measure forecasts beat a well-specified GARCH for daily equity variance, out of sample?
  - does the answer survive once the number of models compared is accounted for?
- Literature
  - Christensen, Siggaard & Veliyev (2023): machine-learning methods with little tuning are competitive with HAR-type models
  - Hansen, Lunde & Nason (2011): the Model Confidence Set is the right object when many models compete
- Formal claims, $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$, each with its own null and rule
  - Equality of A and B: $H_0 : \mathbb{E}d_t = 0$, rejected at 5% if $|t_{DM}| > 1.96$
  - Superiority of A: $H_0 : \mathbb{E}d_t \geq 0$, rejected at 5% if $t_{DM} < -1.645$
  - Membership of the Model Confidence Set concerns the whole set of models, not the pair A–B
  - Stability: a separate test of equal $\mathbb{E}d_t$ across pre-registered subperiods

Open Questions and the Discovery Loop (2/2)

- The loop
  - An Artificial Intelligence (AI) assistant drafts the literature map, competing hypotheses, code, a robustness plan and a referee-style critique
  - The researcher checks every Digital Object Identifier (DOI, a permanent publication identifier), reproduces every number and fixes the plan before the first run
- Checks specific to volatility forecasts
  - Look-ahead: h_t must not move when r_t changes
  - A proxy-robust loss (QLIKE, MSE)
  - Model mining: Model Confidence Set or a Bonferroni correction
  - Scale and convergence of the optimiser
  - DOIs or `arch` function arguments invented by the assistant (reusable prompts and the full checklist: Appendix)

Mini-Case and a Team-Project Seed

- A typical assistant claim: "GJR-t beats GARCH-t because of the leverage effect"
  - d_t = QLIKE(GJR-t) − QLIKE(GARCH-t), 2 693 days, 2016–2026: $\bar{d} = -0.025$, DM $t = -1.22$
- An auxiliary hypothesis, not a necessary implication of leverage: the gain sits after down days
  - Regression on the full calendar: $d_t = a + b\,1(r_{t-1} < 0) + u_t$, HAC with 5 lags
  - $\hat{b} = 0.019$ (SE 0.041, $t = 0.46$): no evidence of a difference between days after a fall (1 223) and the others (1 470)
  - By period: 2016–2020 $t = -0.02$, 2021–2026 $t = -4.06$
  - Descriptive only (nested models, expanding window); calibrated follow-up: the same regression in a fixed-window GW test
- What the result supports: comparison of procedures
- **Project seed**
  - Do machine-learning forecasts enter the Model Confidence Set together with GARCH, GJR and EGARCH for the S&P 500 and the BET, 2016–2026?
  - Replicate the QLIKE table first; pre-register models, losses, splits and error rate
  - Log AI use in `AI_USE.md` and every caught error in `AI_ERRORS.md` (full brief: ▸ Appendix)

Synthesis: a Decision Workflow

- **1. Detect**
  - $Q(10)$ on r_t^2 and the LM test
  - Clustering appears in all six markets; persistence is close to 1, and for Bitcoin and EUR/RON the fit hits the IGARCH bound
- **2. Fit**
  - GARCH, GJR, EGARCH with Normal, Student-t, skewed-t innovations; returns in %
  - Check convergence and parameters at a bound
- **3. Diagnose**
  - Level, squares and signs of $\hat{z}_t$, and its largest values
  - Naive tests are conservative
- **4. Forecast**
  - The recursion gives multi-step variances
  - The square-root-of-time rule ignores mean reversion
- **5. Compare**
  - Out-of-sample: QLIKE, GW with a rolling window and the Model Confidence Set
  - GARCH beats EWMA and Hist-20; the GARCH family is hard to separate
- **6. Report uncertainty**
  - Robust SE (valid if $\mathbb{E}z_t^4 < \infty$) and boundary-aware tests
  - Half-lives as intervals $[h_{lo}; \infty)$ when $\alpha + \beta \approx 1$

Check Yourself, and Next: Chapter 6

Retrieval questions

- Why does the square-root-of-time rule overstate the 10-day risk after a turbulent day?
- Why is the LR test of Normal against Student-t innovations compared with 2.71 and not 3.84?
- Why can a symmetric interval for the half-life contain negative values?
- Why is a gap between the VIX and GARCH volatility not a measured variance risk premium?

Next chapters

- Chapter 6: multivariate volatility and dynamic correlations (DCC)
- Chapters 7–8: Value at Risk and expected shortfall with GARCH volatility, and their backtesting
- Chapter 9: realised volatility and long memory
- Seminar 5: the variance recursion step by step
  - GARCH-t and GJR on real data
  - An AI answer to audit
  - Bootstrap tests and forecast comparisons

Check Yourself: Short Answers

- **Square-root-of-time rule**
  - After a turbulent day $\sigma^2_{t+1} > \sigma^2$, the long-run variance; the daily forecasts decay towards σ^2, so their sum is below $H\sigma^2_{t+1}$
  - Seminar 5, A1
- **2.71 instead of 3.84**
  - Under H_0, $1/\nu = 0$ is on the boundary: in about half of the samples the estimate sits at the bound and $LR = 0$, so the law is $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, whose 5% critical value is 2.71
- **Negative half-lives**
  - $h = \ln 0.5 / \ln p$ explodes as $p \to 1$; the delta method replaces it by a straight line, so $\hat{h} \pm 1.96\,\widehat{\mathrm{SE}}$ ignores the curvature; map the interval for p instead
  - Seminar 5, A2
- **VIX against GARCH**
  - The right comparison: VIX^2, the 30-day risk-neutral expectation of variance, against a 30-day GARCH variance forecast under the real-world probability; $\hat{\sigma}_t$ is a one-day volatility, so horizon and probability both differ

The Practical Decision

- **What I choose**: a GARCH-family model with Student-t innovations; for equity indices, GJR-t
  - GJR-t has the lowest out-of-sample QLIKE (0.731) and an MCS p-value of 1.000
  - EWMA and Hist-20 are clearly worse (DM $t = 3.47$ and 4.10)
- **Under which conditions**
  - The GJR-t persistence is below 1: $\hat{\alpha} + \hat{\gamma}/2 + \hat{\beta} = 0.9835$ for the S&P 500, 0.9897 for the BET
  - Active restrictions are reported: for the S&P 500, $\hat{\alpha} = 0$ hits the bound $\alpha \geq 0$, so the standard error of $\hat{\alpha}$ is not the usual one
  - Student-t MLE standard errors assume the correct density; otherwise robust errors require finite moments of the score, which $\hat{\nu} > 4$ alone does not guarantee
  - $Q(10)$ on $\hat{z}_t^2$ and LM(5) detect no clustering left
- **What result would make me drop it**
  - The estimate hits the bound $\hat{\alpha} + \hat{\beta} = 1$, as for Bitcoin and EUR/RON: the variance forecast then grows by $\hat{\omega}$ per day
  - The model leaves the 90% Model Confidence Set, or loses to EWMA in QLIKE out of sample

Gaussian QMLE: Asymptotic Normality and Standard Errors

- **Asymptotic normality** of $\vartheta = (\omega, \alpha, \beta)$ (pure GARCH, known mean): add ϑ_0 interior and $\kappa_z = \mathbb{E}z_t^4 < \infty$
  - $\sqrt{n}(\hat{\vartheta} - \vartheta_0) \to N\left(0, (\kappa_z - 1)J^{-1}\right)$, $J = \mathbb{E}[g_t g_t']$, $g_t = \partial \ln \sigma_t^2 / \partial \vartheta$
  - κ_z: kurtosis of z_t (3 if Normal); g_t: gradient of $\ln \sigma_t^2$; J: its mean outer product; interior: no constraint binds
- Sandwich $\hat{A}^{-1}\hat{B}\hat{A}^{-1}$ estimates $(\kappa_z - 1)J^{-1}$
  - The inverse Hessian $\hat{A}^{-1}$ estimates $2J^{-1}$
  - A: minus the expected Hessian (second derivatives) of ℓ/n; B: variance of the score (first derivatives) per observation
  - Robust / classical standard error (SE) $= \sqrt{(\kappa_z - 1)/2}$: 1 under Normality, above 1 with heavy tails
- Variants (estimated mean, IGARCH, explosive case): Appendix

QMLE in Our Data: When the Theory Holds and When It Fails

- S&P 500, GARCH(1,1) with Normal likelihood: residual kurtosis $\hat{\kappa}_z = 4.73$
  - Theory: robust / classical SE $= \sqrt{(4.73-1)/2} = 1.37$ (Seminar 5, A5); observed: 1.37 for $\hat{\alpha}$, 1.37 for $\hat{\beta}$
- Bitcoin: GARCH-t gives $\hat{\nu} = 3.18 < 4$ degrees of freedom
  - Under that law $\mathbb{E}z_t^4 = \infty$
  - The condition for Gaussian-QMLE normality fails
  - Gaussian QMLE residual kurtosis does not settle: 19.8 on the first half, 14.6 on the full sample
  - Full sample, robust / classical SE: 2.51 for $\hat{\alpha}$, 2.01 for $\hat{\beta}$, against the formula value 2.61, which assumes $\kappa_z < \infty$
- Hall & Yao (2003): then the Gaussian QMLE converges slower than $\sqrt{n}$
  - To a non-Gaussian limit
  - Its robust t-statistics are not valid
- Remedies
  - A heavy-tailed quasi-likelihood with its scale adjustment Fan, Qi & Xiu (2014)
  - Or the Student-t MLE (bounded score: no fourth moment needed if the law is right)
  - With bootstrap inference near bounds

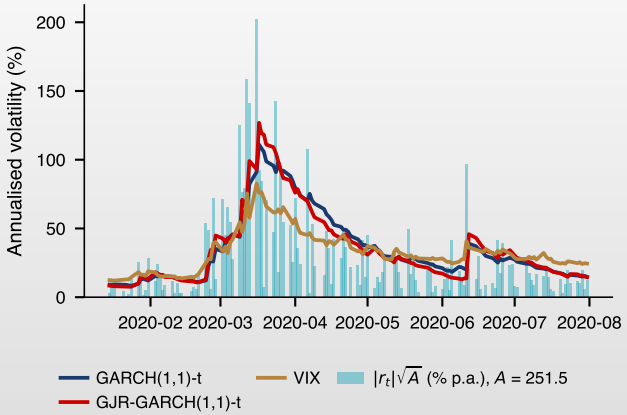
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Boundary Tests on Our Data

- Normal vs Student-t innovations, GARCH(1,1)
  - $LR = 311.9$ (S&P 500), 1 082.0 (Bitcoin; Seminar 5, B6)
  - Far above 2.71: heavy tails are decisive; the mixture matters only in borderline cases
- IGARCH, $H_0: \alpha + \beta = 1$ (GARCH-t): $LR = 1.09$ (S&P 500), 1.14 (BET)
  - S&P 500, p-value: 0.30 (χ^2_1), 0.15 (mixture), 0.17 (parametric bootstrap under H_0 on $\hat{\alpha} + \hat{\beta}$, 998 converged re-estimates of 999 simulated samples)
  - BET, p-value: 0.28, 0.14 and 0.17 (Seminar 5, B8)
- S&P 500, bootstrap under H_0: 53% of the re-estimates have $\hat{\alpha} + \hat{\beta} \geq 0.999$
  - At the bound $\alpha + \beta = 1$, as the mixture predicts
  - Conclusion: a unit root in variance cannot be rejected for the S&P 500 or the BET

MFM_ch5_seminar

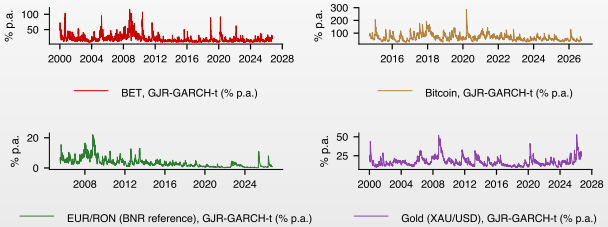
Zoom: The COVID-19 Crash, 2020



- COVID-19, January–July 2020: $|r_t|$ annualised, GARCH-t and GJR-GARCH-t volatility, VIX
- 24 February ($r_t = -3.41\%$): the VIX went from 17.1% to 25.0% the same day
  - GARCH-t from 12.2% to 22.5% only the next day
  - GARCH reacts, markets price at once
- After $r_t = -12.8\%$ on 16 March, peaks on 17 March: GARCH-t 111%, GJR-t 127%
  - The GJR peak is higher because the shocks were negative
  - Full-sample (2000–2026) parameters
  - A forecast with parameters estimated before 2020 (as in Seminar 5, B5) gives a different peak

MFM_ch5_sp500_vix

BET, Bitcoin, EUR/RON and Gold



- BET: 2008 dominates, now 26.4%
  - Bitcoin: 50–100% is normal, the highest peak is March 2020
- EUR/RON: a managed calm with bursts in 2022 and 2025, poorly described by a daily GARCH
  - Gold: a new high of 53% in 2026
- Bitcoin and EUR/RON: the fit is on the IGARCH bound, so the paths are descriptive only (term structure)

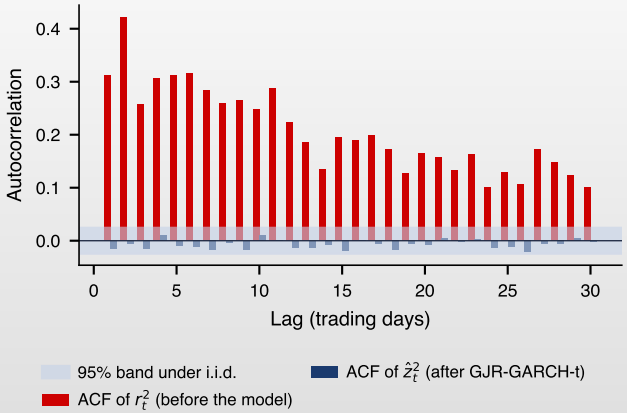
MFM_ch5_markets

Residual Diagnostics Have Estimated Parameters

- $\hat{z}_t$ depends on $\hat{\theta}$: the Ljung–Box statistic on $\hat{z}_t^2$ is not exactly χ^2_m Li & Mak (1994)
  - The fitted variance has already absorbed part of the dependence at small lags
- S&P 500, Gaussian QMLE of GARCH(1,1), $Q(10)$ on $\hat{z}_t^2$
  - Naive Ljung–Box: 16.2, p-value 0.094: not rejected at 5%
  - Corrected for the estimated parameters: 19.0, p-value 0.041: rejected at 5%, the verdict flips
- The naive test is conservative
  - Under H_0 it rejects less often than the nominal level, so it also has less power
  - The joint sign-bias test carries the same caveat (the correction matrix: Appendix)

MFM_ch5_inference

Diagnostics: Squared Returns versus Squared Residuals



- S&P 500, lags 1–30: autocorrelations of r_t^2 (before the model) and of $\hat{z}_t^2$ from GJR-GARCH-t
- After the model they sit inside the i.i.d. band
  - No remaining variance dynamics are detected at these lags
- The band is the naive one: with estimated parameters it is too wide at small lags (previous slide)

MFM_ch5_innovations

Diagnostics in Other Markets: a Decision Table

Market	Residual pattern	Possible failure	Next check
BET	$Q(10)$ on $\hat{z} = 116.6$	autocorrelation in the mean (thin trading, Chapter 2)	AutoRegressive model AR(1) for the mean
Bitcoin	$Q(10)$ on $\hat{z}^2 = 4.7$, joint sign bias 1.3: no rejection	$\hat{\nu} = 3.18 < 4$: $\mathrm{Var}(z_t^2) = \infty$, so the χ^2 calibration is not justified	tests on $\lvert\hat{z}_t\rvert$
EUR/RON	excess kurtosis 3 190	one residual of 154 (May 2025) dominates	inspect the largest residuals; a bounded-score model
Gold	EGARCH: $Q(10)$ on $\hat{z}^2 = 48.6$	clustering left despite a good BIC	choose by diagnostics, not by BIC alone

- Always look at the largest residuals: one observation can dominate every diagnostic

MFM_ch5_innovations

Comparing Forecasting Methods (1/3): the Limit of DM and the Giacomini–White Test

- **The question**: how do we compare forecasts of nested models with estimated parameters?
- **The limit of the DM test** West (1996), Clark & McCracken (2001)
  - With estimated parameters, nested models and an expanding window, the loss differential degenerates under H_0
  - DM is no longer $N(0,1)$
- **The Giacomini–White test** Giacomini & White (2006) tests the forecasting **method**
  - The model plus a rolling window of $W = 1\,000$ days
- The estimation noise does not vanish, so the test also applies to nested models
  - Under the conditions of the theorem: finite moments, limited dependence, a non-singular Ω
- $GW = n\,\bar{Z}^\top \hat{\Omega}^{-1} \bar{Z} \to \chi^2_2$ under $H_0 : \mathbb{E}[d_{t+1} \mid \mathcal{F}_t] = 0$, with $Z_{t+1} = (1, d_t)^\top d_{t+1}$
  - d_{t+1}: loss differential; $\bar{Z}$: mean of Z_{t+1}; $\hat{\Omega} = n^{-1} \sum Z_{t+1} Z_{t+1}^\top$: the uncentred second moment; n: number of forecasts
- Reject at 5% if $GW > 5.99$

Comparing Forecasting Methods (2/3): the Model Confidence Set

- **The question**: which models remain candidates for "the best" when we compare them all?
- With six models, the comparisons with GARCH-t are five tests
  - The **Model Confidence Set (MCS)** Hansen, Lunde & Nason (2011) compares all 15 pairs
- Multiple tests raise the rate of false rejections
- The MCS drops the worst model until equal accuracy is no longer rejected
- The final set contains the best model with probability $\geq 1 - \alpha$, here $\alpha = 10\%$
- Stationary bootstrap: resamples blocks of random (geometric) length
  - Mean 10 days, so that serial dependence is kept

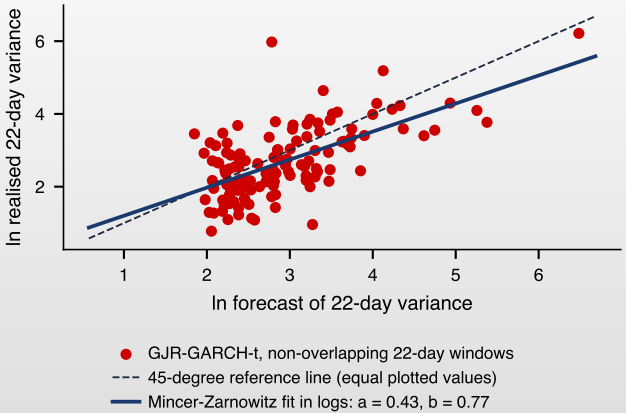
Comparing Forecasting Methods (3/3): Results for the S&P 500

- **Giacomini–White**: rolling window, 2 693 forecasts, each model against GARCH-t
  - Rejects only EWMA and Hist-20
- **Model Confidence Set (MCS) 90%**: expanding window, QLIKE, stationary bootstrap
  - Keeps the four GARCH models, eliminates EWMA and Hist-20
- Same conclusion as the DM test
  - GW, with a rolling window, has calibrated inference under the conditions of the theorem
  - The MCS (expanding window) carries the same caveat as DM
  - Full table and the MCS statistic: Appendix

Model	GW p	MCS p
GARCH-N	0.058	0.588
GARCH-t	–	0.341
GJR-t	0.285	1.000
EGARCH-t	0.202	0.588
EWMA	0.004	0.008
Hist-20	< 0.001	0.002

MFM_ch5_inference

One-Month Forecasts: Mincer–Zarnowitz in Logs (1/2)



- GJR-t forecast of the 22-day variance against the realised sum of 22 squared returns
  - 122 non-overlapping windows
  - Regression $\ln Y = a + b \ln \hat{h} + u$, Y: realised sum, $\hat{h}$: forecast
  - R^2 in logs 0.39
- Slope in logs $b = 0.77$ (SE 0.07): descriptive only
  - Since $\mathbb{E}[\ln Y \mid \mathcal{F}] \neq \ln \mathbb{E}[Y \mid \mathcal{F}]$
  - $b < 1$ does not prove over-dispersed forecasts

MFM_ch5_forecast_evaluation

One-Month Forecasts: Mincer–Zarnowitz in Logs (2/2)

- In levels b is only 0.66: two COVID-19 windows dominate
- Hence we compare the 22-day forecasts with QLIKE Patton (2011)
  - GARCH-N 4.12, GARCH-t 4.13, GJR-t 4.14, EWMA 4.26 (lower = better)
- At one month, the GJR asymmetry no longer pays off

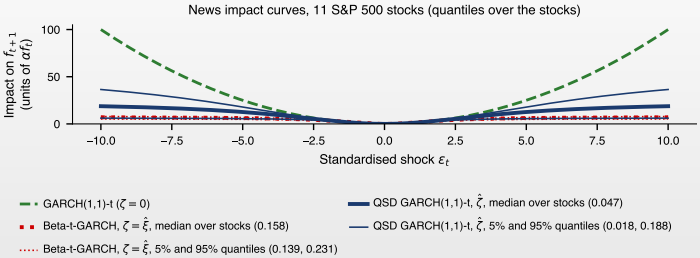
MFM_ch5_forecast_evaluation

QSD: Course Sample and Results

- Course sample: 11 large stocks (the paper: 400)
  - 1995 (or later) – February 2019, 4986–6081 returns each
  - $X_t = \mathrm{VIX}_t^2/252$
- $\zeta = 0$ rejected for 9 of 11 (paper: 89.5%)
  - $\xi = \zeta$ rejected for 8 of 11 (paper: 53.5%)
- Median $1/\hat{\zeta}$: 21.4 (QSD) against 6.32 (Beta-t-GARCH)
  - paper: 9.48 against 5.07
  - 11 selected survivors: imprecise and not representative

MFM_ch5_quasi_score_driven

QSD: News Impact Curves (cf. Paper, Fig. 3)



- Updating functions at quantiles of the 11 estimates, at a common αf_t
  - The contribution of ϵ_t to f_{t+1}, in units of αf_t
  - GARCH-t adds ϵ_t^2 αf_t
- A shock of 5 standard deviations adds 25 (GARCH-t), 12.6 (median QSD), 6.2 (median Beta-t-GARCH)
  - At 10 standard deviations: 100, 18.8, 7.0 (bound $(1+\zeta)/\zeta = 7.3$)
- The QSD downweighting lies between GARCH-t and Beta-t-GARCH

MFM_ch5_quasi_score_driven

Data Used in This Chapter

Series	Owner / source	Start	Obs.	Obs. per year	Vol. (% p.a.)	Excess kurt.
S&P 500	S&P Dow Jones Indices	2000	6 717	251.5	19.2	10.7
BET	Bucharest Stock Exchange	2000	6 670	249.8	22.1	11.1
BET-TR	Bucharest Stock Exchange	2014	2 992	249.7	15.4	17.8
Bitcoin (BTC/USD)	crypto exchanges (composite)	2014	4 384	365.3	66.6	12.0
EUR/RON reference rate	BNR	2005	5 352	252.4	4.4	15.1
Gold (XAU/USD)	spot market	2000	6 949	260.2	17.6	7.7

- Equity indices and gold: weekdays only
- Vol.: sample volatility annualised with each series' observations per year

MFM_ch5_arch_effects

Back

One-Sided Score Tests and the Davies Problem

- The alternative is one-sided ($\alpha_i \geq 0$): the two-sided χ^2 test wastes power
  - Lee & King (1993): one-sided score test built on the sum of the score components, $N(0,1)$ under H_0, rejects in the upper tail only
- Against GARCH(1,1) the null $\alpha = 0$ leaves β unidentified (Davies problem)
  - The ARCH(q) LM statistic is also the score test against GARCH(p,q): the same statistic serves both alternatives (Lee & King (1993))

Back

Robert Engle and the 2003 Nobel Prize



Robert F. Engle (2022)

Photo: Jon Demske (2022); CC BY-SA 4.0; Wikimedia Commons



Clive W.J. Granger (2008)

Photo: Olaf Storbeck (2008); CC BY-SA 2.0; Wikimedia Commons

- Nobel Prize 2003
  - Engle "for methods of analyzing economic time series with time-varying volatility (ARCH)"
  - Shared with Clive Granger (cointegration)
- Nobel lecture: Engle (2004), on risk and volatility in financial practice
- Tim Bollerslev, Engle's doctoral student, generalised ARCH to GARCH in 1986

Back

Kurtosis: Rounding and a Long Simulated Path

- S&P 500, GARCH-N: the moment condition is 0.990509, so $K = 12.2$ has the denominator $1 - 0.990509$
  - Rounding to $\hat{\alpha} = 0.120$, $\hat{\beta} = 0.860$ gives 0.9892 and $K = 11.0$: report such functions from unrounded estimates
- Simulated GARCH(1,1) of the simulation slide, one million days
  - Excess kurtosis 2.47 (theory 3.06); lag-1 autocorrelation of ε_t^2: 0.26
  - The sample kurtosis converges slowly: its precision depends on the eighth moment, infinite for these parameters

Back

QMLE: Variants of the Theorem

- Estimated mean: the score gains the term $\varepsilon_t \, (\partial \mu_t / \partial \theta) / \sigma_t^2$
- The covariance is no longer $(\kappa_z - 1) J^{-1}$: the mean block also depends on the skewness of z_t
- IGARCH ($\alpha + \beta = 1$): strict stationarity still holds Lumsdaine (1996)
  - The QMLE remains consistent and asymptotically Normal
- Explosive case, $\mathbb{E} \ln(\alpha z_t^2 + \beta) > 0$
  - Jensen & Rahbek (2004) estimate (α, β) with ω fixed, since ω cannot be estimated consistently

Back

The Li–Mak Correction for Squared-Residual Autocorrelations

- Squared-residual autocorrelations $\hat{r}_k$: $\sqrt{n}\,\hat{r} \to N(0, C)$, $C = I - H' J^{-1} H/(\kappa_z - 1)$ Li & Mak (1994)
  - $\hat{r} = (\hat{r}_1, \ldots, \hat{r}_m)'$; I: identity matrix; C: the corrected covariance, smaller than I on the diagonal; g_t, J: as in the QMLE theorem
  - $H_k = \mathbb{E}[g_t(z_{t-k}^2 - 1)]$ (k-th column of H): how the fitted variance absorbs lag-k dependence; the ARMA rule χ^2_{m-p-q} does not apply
  - The mean μ is estimated too: C comes from the joint covariance of $\hat{r}$ and the QMLE score for $(\mu, \omega, \alpha, \beta)$, including the cross terms with $\mathbb{E}z_t^3$
- S&P 500, Gaussian QMLE of GARCH(1,1): $C_{1,1} = 0.78$, $C_{5,5} = 0.93$, $C_{10,10} = 0.97$
  - The correction is largest at small lags, where $\hat{\alpha}$ has already fitted the dependence
  - The corrected statistic $n\hat{r}' C^{-1} \hat{r}$ is χ^2_{10} under H_0

MFM_ch5_inference

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S&P 500 Forecast Evaluation, All Statistics

Model	QLIKE	MSE	DM t vs GARCH-t	MZ b	MZ R^2	Wald p
GARCH-N	0.750	22.3	−1.47	0.97	0.250	1.00
GARCH-t	0.756	22.8	–	0.88	0.238	0.80
GJR-t	0.731	21.8	−1.22	0.84	0.276	0.68
EGARCH-t	0.745	21.3	−0.36	1.24	0.293	0.71
EWMA	0.814	24.7	3.47	0.89	0.172	0.69
Hist-20	0.888	26.1	4.10	0.68	0.152	0.03

- Mean losses on 2 693 days
  - DM on QLIKE, HAC with 5 lags
  - Negative DM t: the model beats GARCH-t
- MZ: $r_t^2 = a + b\,h_t$; Wald p: test of $a = 0$, $b = 1$

MFM_ch5_forecast_evaluation

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Giacomini–White and the MCS: Full Table and Settings

Model	QLIKE (W)	t uncond.	GW	p	MCS p
GARCH-N	0.750	1.86	5.70	0.058	0.588
GARCH-t	0.743	–	–	–	0.341
GJR-t	0.737	−0.19	2.51	0.285	1.000
EGARCH-t	0.745	0.06	3.19	0.202	0.588
EWMA	0.814	3.19	11.31	0.004	0.008
Hist-20	0.888	3.87	16.35	< 0.001	0.002

- GW: the statistic of the main text, χ^2_2 under H_0 (5.99 at 5%)
  - t uncond.: the unconditional test (instrument 1 only)
  - A Diebold–Mariano t on the rolling-window losses
- Rolling window $W = 1\,000$ days, re-estimated each January, 2 693 forecasts
  - Tests against GARCH-t: unconditional t uses HAC with 5 lags
  - GW uses the uncentred second moment, without HAC
- MCS on the expanding-window forecasts: QLIKE
  - Statistic T_R: the largest pairwise $|t|$ of loss differences
  - Stationary bootstrap (mean block 10, 10 000 draws)

MFM_ch5_inference Back

Score-Driven, Realized and Component Models

- **Score-driven (GAS)** Creal, Koopman & Lucas (2013): $\sigma^2_{t+1} = \omega + \alpha\, s_t + \beta\, \sigma^2_t$
  - s_t: the scaled score of the Student-t density with respect to σ^2_t
  - The score is bounded in z_t: an outlier such as the EUR/RON residual of 154 moves the variance far less than in GARCH
- **Realized GARCH** Hansen, Huang & Shek (2012): adds a measurement equation linking σ^2_t to a realised variance
  - Uses intraday information and reacts faster; the bridge to Chapter 9
- **GARCH-MIDAS** Engle, Ghysels & Sohn (2013): $\sigma^2_t = \tau_t\, g_t$
  - A slow long-run component τ_t driven by macro data (mixed frequencies)
  - g_t: a unit-mean GARCH(1,1) component for the short-run moves
  - A moving long-run level absorbs the level shifts that push $\hat{\alpha} + \hat{\beta}$ towards 1

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Case Study, Full Method (1/2)

- **Why it matters**: GARCH is driven by ε_t^2 whatever the density
  - So an isolated extreme shock can produce an excessive volatility response
  - The response lasts days or weeks (Section 1 of the paper)
  - With a left-skewed density, a pure score-driven model raises volatility more after positive returns: the opposite of the leverage effect (Example 2)
  - QSD models, eq. (3): $f_{t+1} = \omega + \alpha\,\psi(y_t, X_t, f_t, \theta) + \beta f_t$, ψ the scaled score of an updating function ρ that need not equal $\log p$; theory: consistency, asymptotic normality, LR and LM tests with χ^2 limits (Section 3)
- **Estimation**: $\zeta = 0$ is kept inside the parameter space by eqs. (31)–(32)
  - $\zeta \in (-1, 1/2)$
  - $\Psi(x) = x\,(1 - e^{-cx})/(1 + e^{-cx})$ applied to the weight, $c = 1000$
  - PIT test Diebold, Gunther & Tay (1998) with the Pierce (1982) correction for parameters estimated by maximum likelihood: reject if $|\sum_t \hat{u}_t - T/2| / \sqrt{T(1/12 - B\,\mathcal{I}(\theta)^{-1}B^\top)} > 1.96$
  - $B = \lim_{T\to\infty} T^{-1}\mathbb{E}[\partial \sum_t u_t / \partial\theta^\top]$, $\mathcal{I}$: the Fisher information per observation, T: the sample length; it detects a shifted PIT mean, not every departure from uniformity

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Case Study, Full Method (2/2)

- **Data**: 400 stocks (Section 5)
  - Stocks with fewer than 4000 observations or with convergence problems are dropped
  - Skewed variant, eq. (47): a skewed Student-t with asymmetry κ for ϵ_t and separate (ζ, τ) in the update; $\tau < 0.5$ gives a leverage effect whatever κ (Section 5.2)

Case Study, an Example and the Limits Stated by the Authors

- Centene, 4330 days: $\hat{\xi} \approx 0.26$ (about 4 degrees of freedom) in all three models
  - But $\hat{\zeta} = 0.048$ (SE 0.027)
  - LR 8.81 ($p = 0.003$) for $\xi = \zeta$ and 8.42 ($p = 0.004$) for $\zeta = 0$ (Table 5)
  - After a crash day in March–April 2008, GARCH-t volatility stays well above Beta-t-GARCH
  - QSD lies between (Fig. 4)
- **Limits** stated by the authors
  - Beating GARCH-t and Beta-t-GARCH does not make the Student-t QSD model adequate: the three models fail the PIT test for 40.0–46.0% of the stocks (Table 4)
  - In the unmodified recursion (30), $\zeta < 0$ or $\zeta > 1/2$ does not define a proper volatility model, and $\zeta = 1/2$ with $\varpi = 0$ is degenerate (Section 4.1)
  - The $N(T/2, T/12)$ law of $\sum_t u_t$ holds only for known parameters; with estimated ones the PIT test needs the correction above (Section 5.1)
- **Open extensions** (Seminar 5, C3)
  - Does the bounded QSD update beat GJR-GARCH-t in QLIKE out of sample, especially after March 2020?
  - Does it tame the EUR/RON residual of 154?

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The Discovery Loop: Reusable Prompts (1/2)

- Tools are interchangeable
  - An AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot)
  - Literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature**
  - *"List out-of-sample comparisons of GARCH, HAR and machine-learning variance forecasts since 2015, with DOI, asset, loss function, test and horizon."*
  - Resolve every DOI on Crossref; check that the loss is robust to a noisy proxy Patton (2011)
- **Hypotheses**
  - *"For the claim that asymmetry improves one-day forecasts, list competing mechanisms (leverage, volatility feedback, a few crisis days) and what each predicts for d_t given the sign of r_{t-1} and by subperiod."*
- **Data and code**
  - *"Write Python with `arch` that re-estimates GARCH-t and GJR-t each January on an expanding window from the course daily price files and stores one-day variance forecasts; no parameter may use data after the forecast origin."*
  - Replicate first: QLIKE 0.756 (GARCH-t) and 0.731 (GJR-t), DM $t = -1.22$

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The Discovery Loop: Reusable Prompts (2/2)

- **Robustness**
  - *"Write a pre-registered plan: splits by the sign of r_{t-1} and by subperiod, MSE besides QLIKE, the 22-day horizon, the HAC lag, and a Bonferroni or Model Confidence Set correction."*
  - A random-split placebo is calibrated only if it reassigns blocks of days, preserving volatility clustering
- **Critique**
  - *"Act as a referee: could the proxy r_t^2, one crisis year or the HAC lag create my result? Give the test that would expose each."*
- **Report**: all splits and losses tried, adjusted p-values, code, seeds and package versions
  - A specification chosen after seeing results is exploratory and reported as such
- **Influential periods**: a crisis year can dominate $\bar{d}$
  - Report subperiods and state in advance whether any year may be excluded

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The Team-Project Seed in Full

- Realised-variance (HAR) branch: 40 large American stocks
  - The course data set of 5-minute realised variances, January 2015 – January 2026
  - There are no intraday data for the S&P 500 index or the BET, so this branch is run stock by stock
- Replicate first the QLIKE table and the mini-case splits
- Deliverables
  - A one-page pre-registration: models, losses, splits, error rate and what would falsify the claim
  - Code on the course market data, the Model Confidence Set at 10%, a referee-style self-critique; `AI_USE.md` and `AI_ERRORS.md`
- Code of the mini-case: Quantlets/Ch_05/ai_discovery_case.py

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Further Resources

- Books
  - Franke, Härdle & Hafner (2019), chapters on AutoRegressive Integrated Moving Average (ARIMA) and GARCH models for financial time series
  - Francq & Zakoïan (2019): the mathematical theory of GARCH models
- Quantinar courses
  - Statistics of Financial Markets
  - Applied Time Series Analysis with Python
- Code
  - All charts and tables: Quantlets/Ch_05; Python package `arch` by Kevin Sheppard (documentation)
- Related work: entropy as a tail-risk signal beyond variance
  - Pele, Lazar & Dufour (2017): entropy of intraday EUR/JPY returns, Value at Risk and expected shortfall
  - Pele & Mazurencu-Marinescu-Pele (2019): intraday Bitcoin entropy and daily Value at Risk
  - Pele, Lazar & Mazurencu-Marinescu-Pele (2019): tail entropy and expected shortfall

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Acronyms used in this material (1/2)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BIC**: Bayesian Information Criterion
- **BTC**: Bitcoin
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **COVID-19**: COronaVIrus Disease 2019
- **DCC**: Dynamic Conditional Correlation
- **DOI**: Digital Object Identifier
- **DM**: Diebold–Mariano (test)
- **EGARCH**: Exponential GARCH
- **EUR**: euro, international currency code
- **EWMA**: Exponentially Weighted Moving Average
- **FIGARCH**: Fractionally Integrated GARCH
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-M**: GARCH-in-Mean
- **GARCH-MIDAS**: GARCH with Mixed Data Sampling (long-run component)
- **GARCH-N**: GARCH with Normal innovations
- **GAS**: Generalised Autoregressive Score (model)

Acronyms used in this material (2/2)

- **GED**: Generalised Error Distribution
- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **GW**: Giacomini–White (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **IGARCH**: Integrated GARCH
- **LM**: Lagrange Multiplier
- **LR**: Likelihood Ratio (test)
- **MCS**: Model Confidence Set
- **MF2-GARCH**: Multiplicative Factor Multi-Frequency GARCH
- **MLE**: Maximum Likelihood Estimation
- **MSE**: Mean Squared Error
- **MZ**: Mincer–Zarnowitz (regression)
- **NIC**: News Impact Curve
- **PIT**: Probability Integral Transform
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QMLE**: Quasi-Maximum Likelihood Estimator
- **QQ**: Quantile–Quantile (plot)
- **QSD**: Quasi Score-Driven (model)
- **RMSE**: Root Mean Squared Error
- **RON**: Romanian leu, international currency code
- **SE**: Standard Error
- **US**: United States
- **USD**: US dollar, international currency code
- **VIX**: Chicago Board Options Exchange Volatility Index
- **XAU**: gold, international currency code (one troy ounce)

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